# Paper 22: Primorial Cumulant Noether Theory and the Riemann Hypothesis

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**Abstract**

We use the all-order cumulant viewpoint opened by Paper 17 and the primorial pattern suggested by Paper 21 to formulate the sharpest current route from the simplicial-cyclotomic program to the Riemann Hypothesis. The central object is the primorial tower

$$P_n = p_n\# = \prod_{i=1}^{n} p_i, \qquad R_n = \mathbb{Z}/P_n\mathbb{Z},$$

whose conductor lattice contains exactly the squarefree interactions among the first $n$ prime axes. The $k$-th Lagrangian/cumulant order naturally probes the row $P_{k-1}$: quadratic order sees the first prime axis, cubic order sees $2 \cdot 3$, with the first nonabelian rung forced by Paper 18, Theorem B, quartic order sees $2 \cdot 3 \cdot 5$, quintic order sees $2 \cdot 3 \cdot 5 \cdot 7$, and the all-order limit sees the natural-number prime-exponent lattice.

The proposed bridge to the Riemann Hypothesis is not a direct zero calculation. It is a positivity theorem. In the language of Paper 20, this is the all-order version of the same boundedness-below requirement: finite projection-closure residuals are nonnegative inside the information geometry, and the limiting cumulant tower should identify that lower bound with the Weil quadratic form. The all-order Noether energy on the primorial critical leaf should converge to the Weil explicit-formula quadratic form for the completed zeta function. Since the Weil positivity criterion is equivalent to the Riemann Hypothesis, this gives a precise target:

| primorial closed-current Noether positivity $\implies$ Weil positivity $\implies$ RH. |
|---|

The paper proves the formal conditional implication, supplies an explicit canonical candidate for the recovery maps $\mathcal{R}_n$, records exact low-order agreement at quadratic and cubic order, and reduces the remaining gap to two finite-place analytic inputs for the von Mangoldt explicit formula, explicit projective-closure hypotheses for the critical leaf, and the Archimedean Noether Limit. Thus this is not advertised as an unconditional proof of RH, and it is not a physical-gravity or spacetime stability claim. It is the best current candidate for turning the simplicial-cyclotomic machinery into one.

**Keywords:** Riemann Hypothesis, primorial, cumulant Noether theorem, information geometry, Weil positivity, explicit formula, Euler product, prime-exponent lattice, Amari-Chentsov tensor, conductor lattice, CRT.

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## 1. The Point of Contact

Paper 5 identified three cautious interfaces with RH:

1. the prime-exponent lattice behind the Euler product;
2. spectral programs of Berry-Keating, Connes, Bost-Connes, and Tate;
3. finite simplicial zeta analogues.

Papers 10 and 11 then supplied an all-order conductor-convolution expansion: the $n$-th Fisher Taylor coefficient is controlled by the $n$-fold convolution $\widehat{h}^{*n}$ in the character group, and conductor-block compression is governed by Hayes-Mobius coefficient vectors in a finite quotient-character adjacency algebra. Paper 17 added the missing variational ingredient: a cubic action-energy theorem and a Noether dictionary at third cumulant order. Paper 21 then exposed the next organizing principle: quartic order points canonically to the first three-prime row 30.

The new synthesis is this:

> The all-order cumulant Noether tower should be indexed by the primorial tower. The primorial rows are the finite squarefree stages of the prime-exponent lattice, and their closed-current Noether energies are the finite arithmetic shadows of the zeta explicit formula.

This is stronger than saying "primes appear." It says that primes are the local tensor axes, while natural numbers are the occupation states generated by those axes. The primorial row $P_n$ is the minimal finite stage on which the first $n$ prime axes can interact simultaneously.

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## 2. Primorial Rows and Lagrangian Order

Let

$$P_n = p_n\# = \prod_{i=1}^{n} p_i, \qquad R_n = \mathbb{Z}/P_n\mathbb{Z}.$$

By the Chinese Remainder Theorem,

$$R_n \cong \prod_{i=1}^{n} \mathbb{Z}/p_i\mathbb{Z}.$$

The nontrivial conductor blocks of the categorical tangent space over $R_n$ are indexed by nonempty subsets of the first $n$ prime axes, equivalently by nontrivial divisors of $P_n$.

This gives the finite squarefree skeleton of the natural numbers. Prime powers do not appear as new axes; they appear as occupations along an existing prime axis in the Euler product:

$$\zeta(s) = \prod_{p} (1 - p^{-s})^{-1} = \sum_{k \in \mathbb{N}^{(\mathcal{P})}} \exp\left( -s \sum_{p} k_p \log p \right).$$

Thus there are two independent cutoffs:

1. **Axis cutoff:** keep primes $p \le p_n$, giving $P_n$.
2. **Occupation cutoff:** bound the exponents $k_p$.

The primorial tower is the canonical axis cutoff. It is the correct object for detecting which cumulant order first forces how many distinct prime axes to interact.

The proposed order rule is

$$\boxed{\ell_k \text{ naturally probes } R_{k-1} = \mathbb{Z}/(p_{k-1}\#)\mathbb{Z}.}$$

Equivalently:

| order | interaction | minimal row | new question |
|---|---|---|---|
| 2 | binary/Fisher | 2 | one prime axis |
| 3 | cubic/Amari-Chentsov | 6 | first conductor interaction |
| 4 | quartic/jerk | 30 | triple conductor mixing |
| 5 | quintic | 210 | coherence among triple faces |
| $k$ | $k$-th cumulant | $p_{k-1}\#$ | $(k-1)$-axis coherence |

This table is the new reason Paper 21 should treat $R = \mathbb{Z}/30\mathbb{Z}$ as canonical rather than merely convenient.

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## 3. The All-Order Cumulant Action

Let $h = \eta - \eta_*$ be displacement from the categorical centroid and $v = \dot{\eta}$. Paper 17 separated the accumulated action from the instantaneous Noether energy:

$$\mathcal{A}_3[\eta] = \int \ell_3(\eta, \dot{\eta})\, dt, \qquad \mathcal{E}_3 = \text{Noether energy density.}$$

At third order,

$$\ell_3(\eta, v) = \frac{1}{2} g_\eta(v, v) + \frac{1}{3} T_\eta(h, v, v),$$

where $T = K_{CH}$ is the Amari-Chentsov tensor.

The all-order continuation should be a cumulant-jet action

$$\mathcal{A}^{(n)}_{\le K}[\eta] = \int \ell^{(n)}_{\le K}(\eta, \dot\eta)\, dt$$

with

$$\ell^{(n)}_{\le K}(\eta, v) = \frac{1}{2} g_\eta(v, v) + \sum_{r=3}^{K} c_r\, T^{(r)}_\eta(h^{\otimes(r-2)}, v, v),$$

where $T^{(3)} = T$, $T^{(4)} = \nabla^{(e)} T$, and the constants $c_r$ depend on the chosen cumulant normalization. The support and Noether questions do not depend on those constants.

Time-translation invariance gives an instantaneous Noether energy density

$$\mathcal{E}^{(n)}_{\le K} = v \cdot \frac{\partial \ell^{(n)}_{\le K}}{\partial v} - \ell^{(n)}_{\le K}.$$

The formal all-order object is the limit

$$\mathcal{A}^{(n)}_{\infty} = \lim_{K\to\infty} \mathcal{A}^{(n)}_{\le K}, \qquad \mathcal{E}^{(n)}_{\infty} = \lim_{K\to\infty} \mathcal{E}^{(n)}_{\le K},$$

whenever the cumulant expansion is interpreted as a formal jet or as a convergent local expansion in the simplex interior.

The point is not that the real simplex dynamics literally terminates at cubic order. It does not. The cubic term is the first non-Gaussian correction. The jerk is the first witness that the cubic theory is a truncation of an infinite cumulant tower.

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## 4. Closed Currents and the Critical Leaf

The RH bridge requires a distinguished limiting subspace. The older critical leaf note proposed the right template:

$$\text{closed currents on Crit} \longleftrightarrow \text{explicit-formula Dirichlet/Euler data.}$$

For each primorial row $R_n$, let $\mathcal{J}_n$ be the finite quotient-character adjacency algebra of Paper 11, and let

$$\mathsf{C}^{(n)}_m(h) = \sum_{\mathfrak{e},\mathfrak{f}} \sum_{\eta} \hat h^{*m}(\eta)\, A^{\mathfrak{f},\mathfrak{e}}_\eta \in \mathcal{J}_n$$

be the order-$m$ Hayes-Mobius coefficient vector.

Define a finite leakage form

$$\Lambda_n(J) \ge 0$$

on admissible currents $J \in \mathcal{J}_n$ by summing the squared non-closed components of the all-order Noether variation:

$$\Lambda_n(J) = \sum_{m\ge 3} w_m \left\| \text{nonclosed}\big(\delta \mathcal{E}^{(n)}_m(J)\big) \right\|^2, \qquad w_m > 0.$$

The precise weights are a normalization choice; the essential feature is positivity.

The finite critical leaf is the minimizing closed-current sector:

$$\mathrm{Crit}_n = \{J : \Lambda_n(J) = \min \Lambda_n\}$$

after quotienting the quadratically flat degeneracies. The tower claim is that the $\mathrm{Crit}_n$ stabilize projectively and that every off-critical leaf has a persistent positive leakage, residual hyper-adjacency obstruction, or nonzero carry class.

This is the all-order Noether version of the critical-line idea:

> The critical line is the analytic shadow of the unique conductor leaf on which the all-order Noether current closes in the primorial tower.

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## 5. The Noether-Weil Quadratic Form

The classical Riemann Hypothesis has a positivity formulation. In one of its standard forms, Weil's explicit formula associates to a suitable test function $F$ a quadratic form $\mathcal{W}_\zeta(F)$. RH is equivalent to positivity of this form on functions of the shape $F = f * f^*$, after the usual completed zeta normalization.

This formulation is used here as a known external theorem, not reproved. The normalization of $\mathcal{W}_\zeta$, the allowed test-function space, and the gamma-factor terms must be fixed in the analytic paper that proves the recovery identity below. At the present stage $\mathcal{W}_\zeta$ denotes the standard completed-zeta Weil form.

The proposed primorial Noether object is a finite positive form

$$\mathcal{Q}_n(J) = \mathcal{E}_\infty^{(n)}(J) + \Lambda_n(J)$$

restricted to currents $J$ on $\mathrm{Crit}_n$. It is positive by construction:

$$\mathcal{Q}_n(J) \geq 0.$$

**Remark 5.0 (Boundedness below at the tower limit).**
This is the all-order analogue of the order-4 story in Paper 20. There the jerk/simplex mass term is a positive projection-closure residual for time-translation energy bookkeeping after diagonal projection. Here the finite leakage forms $\Lambda_n$ and closed-current Noether energies play the same role through the whole cumulant tower: they assert that admissible primorial currents have an information-geometric energy bounded below by zero after the nonclosed components are penalized or projected away. The proposed Noether-Weil bridge is the conditional claim that this bounded-below finite structure survives the projective and Archimedean limits and becomes Weil positivity. It is not a claim about physical gravity, a spacetime Hamiltonian, or an unconditional proof of RH.

*Forward reference (Paper 32 §7 / §12 INSERT 13 status pin).* Paper 32 §7 Theorem 7.1's "gravity-sector stability $= m^2_{\mathrm{simplex}} \geq 0$" reading is the physics-homolog face of the *same* cubic-order shadow recorded here: at the $P_2 = 2 \cdot 3$ row, the cubic-order instance of this positivity is exactly the non-tachyonicity of Paper 20's simplex mass term — the program's order-4 jerk and the all-order primorial leakage forms $\Lambda_n$ are *the same boundedness-below requirement at different cumulant orders*. Both Papers (22 §9.2 and 32 §7) inherit the same [C] all-order status: the cubic match is the confirmed-faithful base-case shadow; the all-order limit is Conjecture 5.2, cornered into §8's

four named classical obligations, none discharged. The two papers' RH language is synchronized so neither can be cited as stronger than the other (Paper 22 §11 "Unsafe statements", binding on both).

The missing analytic theorem is the following identity.

**Definition 5.1 (Primorial recovery current, conductor-cascade form).**
Let

$$R_n = \mathbb{Z}/P_n\mathbb{Z}, \qquad P_n = p_n\#,$$

and let $H_{R_n}$ be its real zero-sum tangent space. Let $Q$ be a conductor-respecting Helmert/NZSBTL cascade on $H_{R_n}$. Equivalently, $Q$ is a packet-refining flag for the CRT conductor packets

$$H_{R_n} = \bigoplus_{\substack{d|P_n \\ d>1}} H_n^{[d]}.$$

Write

$$\Gamma_Q : H_{R_n} \longrightarrow \bigoplus_{\substack{d|P_n \\ d>1}} H_{n,Q}^{[d]}$$

for the NZSBTL coordinate chart extracted from the cascade, $U_Q = \Gamma_Q^{-1}$ for the inverse chart, and $\Pi_d^Q$ for coordinate projection onto the lines of $Q$ lying in the conductor packet $d$. Since $Q$ is conductor-respecting,

$$P_d = U_Q \Pi_d^Q \Gamma_Q$$

is the canonical conductor packet projector on $H_n^{[d]}$, independent of the harmless signed or orthogonal choices made inside that packet.

Let $f \in \mathcal{S}_0$ be a Weil test function, smooth and compactly supported on $\mathbb{R}_{>0}$, whose Mellin transform $\tilde{f}(s)$ decays rapidly in vertical strips. Fix a cutoff $M_n \to \infty$. For each $P_n$-smooth squarefree integer $m \le M_n$, define

$$d(m) = \prod_{p|m} p,$$

and let $\eta_m \in \widehat{R}_n$ denote the additive character label whose primitive CRT support is $d(m)$. Put

$$a_m(f) = \tilde{f}\left(\frac{2\pi i \log m}{\log P_n}\right).$$

The cascade-generated finite Fourier datum is

$$\hat{h}_{n,Q}^f(\eta) = \sum_{\substack{m \le M_n \\ m \ P_n\text{-smooth squarefree}}} a_m(f) \left\langle e_\eta, U_Q \Pi_{d(m)}^Q \Gamma_Q e_{\eta_m} \right\rangle,$$

where $e_\eta$ is the real Fourier character vector. Because $Q$ refines the conductor packets, this reduces to the explicit primorial datum

$$\hat{h}_{n,Q}^f(\eta) = \sum_{\substack{m \le M_n \\ m \ P_n\text{-smooth squarefree} \\ \eta_m = \eta}} \tilde{f}\left(\frac{2\pi i \log m}{\log P_n}\right).$$

Thus the cascade does not add arbitrary coordinates; it variationally selects the conductor chart in which the old character datum is canonical. The primorial Dirichlet cutoff is chosen so that the omitted smooth-number tail is $o(1)$ in the Weil test-function topology.

The raw Hayes-Mobius current generated by $Q$ is the all-order package

$$J^{\mathrm{raw}}_{n,Q}(f) = \Big(\mathsf{C}^{(n)}_k(\hat{h}^{\,f}_{n,Q})\Big)_{k\ge 2},$$

where

$$\mathsf{C}^{(n)}_k(\hat{h}^{\,f}_{n,Q}) = \sum_{\mathfrak{e},\mathfrak{f}}\sum_{\eta}(\hat{h}^{\,f}_{n,Q})^{*k}(\eta)\,A^{\mathfrak{f},\mathfrak{e}}_{\eta} \in \mathscr{J}_n.$$

The CRT tensor decomposition of $\mathscr{J}_n$ sorts this current by prime-support packets. Finally project orthogonally, with respect to the finite leakage form $\Lambda_n$, onto the minimizing closed sector:

$$\mathcal{R}_{n,Q}f = \Pi^{\Lambda}_{\mathrm{Crit}_n} J^{\mathrm{raw}}_{n,Q}(f) \in \mathrm{Crit}_n.$$

If $\mathcal{I}_n$ is the Paper 25 information-investment functional on an admissible conductor-cascade class $\mathcal{A}_n$, choose

$$Q^\star_n \in \operatorname*{arg\,min}_{Q\in\mathcal{A}_n} \mathcal{I}_n(Q)$$

and define the variational recovery map

$$\mathcal{R}^{\mathrm{var}}_n f = \mathcal{R}_{n,Q^\star_n} f.$$

It is canonical once the cutoff $M_n$, the Weil normalization, the leakage inner product, and the variational tie-breaking inside the minimizing sector are fixed. It is functorial under the primorial tower maps after quotienting flat degeneracies, and it reduces to the identity on the quadratic Fisher packet.

**Proposition 5.1.1 (Recovery map generation).**
For every conductor-respecting minimizing cascade $Q^\star_n$, the variational recovery map $\mathcal{R}^{\mathrm{var}}_n$ coincides with the recovery map defined above from the explicit primorial Fourier datum. In particular,

$$\mathcal{R}^{\mathrm{var}}_n f \in \mathrm{Crit}_n$$

for every $f \in \mathcal{S}_0$. Under the hard packet-preserving constraint $Q^\star_n \in O_{\mathrm{pkt}}$, the raw Hayes-Mobius current already lies in the closed sector whenever the leakage form detects exactly cross-packet nonclosure. Under first-order packet pricing, the only remaining defect is the priced carry term measured by the Paper 25 frame-price frontier $\mu^\star_n$.

*Proof.* Since $Q^\star_n$ is conductor-respecting, its coordinate projectors conjugate back to the canonical conductor packet projectors:

$$U_{Q^\star_n}\Pi^{Q^\star_n}_d\Gamma_{Q^\star_n} = P_d.$$

The character vector $e_{\eta_m}$ lies in the primitive packet $H^{[d(m)]}_n$. Hence

$$P_{d(m)}e_{\eta_m} = e_{\eta_m}, \qquad P_d e_{\eta_m} = 0 \quad (d \neq d(m)).$$

Substituting this into the cascade-generated formula gives exactly the finite Fourier datum of the original recovery definition. The Hayes-Mobius coefficient vectors are polynomial convolution expressions in that datum, so the compressed current $J^{\mathrm{raw}}_{n,Q^\star_n}(f)$ is the same element of the quotient-character adjacency algebra as the original raw recovery current, with the CRT tensor decomposition only rewriting it by prime-support factors.

The final projection

$$\Pi^{\Lambda}_{\mathrm{Crit}_n} : \mathcal{J}_n \to \mathrm{Crit}_n$$

lands in $\mathrm{Crit}_n$ by definition of the minimizing closed sector. This proves the membership statement and the equality with the original recovery map. If the admissible class is $O_{\mathrm{pkt}}$, then no cross-packet component is created by the cascade, so any leakage form whose nonclosed part is precisely cross-packet escape vanishes on the raw current. With the first-order packet-respect term instead, the same cross-packet escape is not forbidden but is exactly the quantity priced by $\mu^{\star}_n$. Thus the projected current lies in the same minimizing sector, while the unprojected defect is the priced carry measured by the frame frontier. □

**Theorem 5.1.2 (Leakage control by first-order frame price).**
Fix $f \in \mathcal{S}_0$ and a cutoff $M_n$. Let

$$Q^{\star}_n \in \operatorname*{arg\,min}_{Q \in \mathcal{A}_n} \mathcal{J}_n(Q)$$

be a conductor-respecting minimizer in one of the regularized strata of Paper 25, Theorem 3.8: either the hard packet-preserving stratum $O_{\mathrm{pkt}}$, or a reuse/first-order-priced stratum with $\mu \geq \mu^{\star}_n(\mathcal{H}_n)$.

Transport the Paper 25 first-order packet seminorm $N$ to raw Hayes-Mobius currents by declaring $\varepsilon_n(J)$ to be the smallest first-order packet-escape size of a conductor chart that generates $J$, after quotienting within-packet rotations and flat leakage degeneracies. In the frame-generated sector this is equivalently the supremum of the normalized cross-packet first-order Noether variation:

$$\varepsilon_n(J) = \sup_{\nu(A)>0} \frac{|a_J(A)|}{\nu(A)}.$$

Assume the finite leakage form is normalized so that its nonclosed component is Lipschitz, on the Hayes-Mobius image of the cutoff test-function ball, with respect to this transported first-order packet-escape seminorm:

$$\Lambda_n(J) \leq L_f(n)\, \varepsilon_n(J),$$

where

$$L_f(n) \leq C_f \ \mathrm{polylog}(P_n).$$

Then the variational recovery current satisfies

$$\Lambda_n(\mathcal{R}^{\mathrm{var}}_n f) \leq C_f\, \mu^{\star}_n(\mathcal{H}_n)\, \mathrm{polylog}(P_n).$$

In the hard packet-preserving stratum, under the same normalization,

$$\Lambda_n(\mathcal{R}^{\mathrm{var}}_n f) = 0.$$

Consequently, if along the primorial tower the reuse frontier is regularized as in Paper 25, Theorem 3.8, and

$$\mu^{\star}_n(\mathcal{H}_n)\, \mathrm{polylog}(P_n) \longrightarrow 0,$$

then

$$\Lambda_n(\mathcal{R}^{\mathrm{var}}_n f) \longrightarrow 0.$$

*Proof.* The leakage form is nonnegative and $\mathrm{Crit}_n$ is its minimizing closed sector. By Proposition 5.1.1, the recovery current is obtained from the cascade-generated raw current by the orthogonal leakage projection

$$\mathcal{R}^{\mathrm{var}}_n f = \Pi^{\Lambda}_{\mathrm{Crit}_n} J^{\mathrm{raw}}_{n,Q^{\star}_n}(f).$$

Projection to a minimizing sector cannot increase the leakage distance to that sector; equivalently, after quotienting the flat degeneracies used in the definition of $\mathrm{Crit}_n$,

$$\Lambda_n(\mathcal{R}_n^{\mathrm{var}} f) \le \Lambda_n(J^{\mathrm{raw}}_{n,Q_n^\star}(f)).$$

In the hard packet-preserving stratum, $Q_n^\star \in O_{\mathrm{pkt}}$. Paper 25, Theorem 3.6 removes all admissible cross-packet escape directions: the cascade preserves every conductor packet and $\varepsilon_n(J^{\mathrm{raw}}_{n,Q_n^\star}(f)) = 0$. Since the present leakage normalization detects exactly the nonclosed cross-packet component on the Hayes-Mobius image, the raw current is already closed, and therefore the displayed leakage is 0.

In the first-order-priced stratum, Paper 25, Theorem 3.8(S3) says that the only possible infinitesimal top-prefix escape is controlled by

$$\mu_n^\star(\mathcal{H}_n) = \sup_{\nu(A)>0} \frac{|a_{\mathcal{H}_n}(A)|}{\nu(A)}.$$

Theorem 3.8(S2) gives the same statement with $a_{\mathcal{H}_n}$ replacing one-point derivatives on reuse families. Thus every first-order nonclosed component of the frame-generated Hayes-Mobius current is bounded, after normalization by the packet seminorm $\nu(A)$, by the critical price:

$$\varepsilon_n(J^{\mathrm{raw}}_{n,Q_n^\star}(f)) \le \mu_n^\star(\mathcal{H}_n).$$

Applying the Lipschitz comparison between Noether leakage and transported packet escape,

$$\Lambda_n(J^{\mathrm{raw}}_{n,Q_n^\star}(f)) \le L_f(n)\varepsilon_n(J^{\mathrm{raw}}_{n,Q_n^\star}(f)) \le C_f\,\mu_n^\star(\mathcal{H}_n)\,\mathrm{polylog}(P_n).$$

Combining this with the projection inequality proves the stated bound.

The projective-limit statement follows immediately from the same inequality. The packet-local laws of Paper 25, Theorems 3.10–3.14, justify using $\mu_n^\star$ as a controlled tower quantity: fixed packets are stable under deeper local embeddings, top packets have the stated closed-form decay laws, and odd-residue realification changes the envelope only by the explicit factor already included in the normalization. Thus the remaining primorial input is the squarefree/CRT-face analogue of that packet-local control; once the displayed product tends to zero, leakage extinction follows. □

**Definition 5.1.3a (squarefree/CRT restriction package).**
Let $S$ be a finite set of primes and put

$$P_S = \prod_{p\in S} p, \qquad R_S = \prod_{p\in S} \mathbb{F}_p \cong \mathbb{Z}/P_S\mathbb{Z}.$$

For a CRT additive-character label $\eta = (\eta_p)_{p\in S}$, define

$$\mathrm{supp}(\eta) = \{p \in S : \eta_p \neq 0\}, \qquad d(\eta) = \prod_{p\in \mathrm{supp}(\eta)} p.$$

The real zero-sum tangent space decomposes as

$$H_S = \bigoplus_{\substack{d\mid P_S \\ d>1}} H_S^{[d]},$$

where $H_S^{[d]}$ is the realified additive-character packet of primitive support $d$. For a squarefree face $d \mid P_S$, write

$$V_{|d}^S = \bigoplus_{\substack{e|d \\ e>1}} H_S^{[e]}.$$

If $T \subset S$, let $\rho_{S,T} : R_S \to R_T$ be the CRT projection. The restriction map on normalized character bases is defined as the adjoint of the normalized pullback:

$$\iota_{T,S} = \sqrt{|R_T|/|R_S|}\, \rho_{S,T}^*, \qquad r_{S,T} := \iota_{T,S}^*.$$

Thus

$$r_{S,T} e_\eta^S = \begin{cases} e_{\eta|T}^T, & \eta_{S\setminus T} = 0, \\ 0, & \text{otherwise.} \end{cases}$$

In cyclic labels, when $P_T \mid P_S$, this keeps exactly the labels

$$b = (P_S/P_T)a$$

and sends $b \mapsto a$.

The current restriction

$$\pi_{S,T} : \mathcal{J}_S \longrightarrow \mathcal{J}_T$$

is defined by applying $r_{S,T}$ to every character/tensor leg in the quotient-character adjacency algebra, killing packets with new-prime support, and then quotienting flat leakage degeneracies. The projective-closure program must verify functoriality

$$\pi_{U,T}\pi_{S,U} = \pi_{S,T}, \qquad T \subset U \subset S,$$

on $\mathcal{J}_S/\mathrm{Flat}_S$.

**Criterion 5.1.3b (leakage compatibility and lower semicontinuity).**
Write the finite leakage in the form

$$\Lambda_S(J) = \sum_{k\geq 3} w_k \, \|\mathcal{N}_{S,k}(J)\|^2,$$

where $\mathcal{N}_{S,k}$ is the nonclosed part of the order-$k$ Noether variation. A sufficient squarefree/CRT verification of Theorem 5.1.3, assumptions (1)–(3), is:

1. for each fixed $T \subset S$ and $k$,

$$\mathcal{N}_{T,k}\pi_{S,T} = \pi_{S,T}^{(k)}\mathcal{N}_{S,k} + E_{S,T,k}, \qquad \|E_{S,T,k}\| \to 0$$

   along cofinal $S$ with $T$ fixed;
2. $\pi_{S,T}$ is a contraction in the leakage norm, after quotienting flat degeneracies and using compatible weights $w_k$;
3. the closed sectors $\mathrm{Crit}_S$ are stable under restriction after quotienting flats, and an off-critical class cannot disappear under restriction unless it is already flat.

Then

$$\Lambda_T(\pi_{S,T}J_S) \le \Lambda_S(J_S) + o_{S,T}(1),$$

and the projective leakage

$$\Lambda_\infty(J) = \sup_T \Lambda_T(\pi_{\infty,T}J)$$

is lower semicontinuous as a supremum of finite continuous quadratic forms on the quotient tower.

**Target Lemma 5.1.3c (squarefree CRT-face frame-price stability).**
Let $d \mid P_T \mid P_S$ and let $V_{|d}^S$ be the active CRT face. For a displacement family $\mathcal{H}$ factoring through $R_d$, define the unpriced face ledger $J_{0,S,d}$, the first variation

$$a_{S,d,\mathcal{H}}(A) = \mathbb{E}_{h\in\mathcal{H}} \frac{d}{d\theta} J_{0,S,d}(\exp(\theta A); h)\big|_{\theta=0},$$

and the cross-packet seminorm

$$\nu_S(A) = \sum_{e\neq f} \|P_e A P_f\|_F.$$

Put

$$\mu^\star_{S,d}(\mathcal{H}) = \sup_{\nu_S(A)>0} \frac{|a_{S,d,\mathcal{H}}(A)|}{\nu_S(A)}.$$

The desired squarefree analogue of Paper 25 packet-local stability is:

$$\mu^\star_{S,d}(\mathcal{H}) = \mu^\star_{d,d}(\mathcal{H})$$

for proper CRT faces, after identifying $V_{|d}^S \cong H_d$. The proof should decompose cross-packet directions into inside-face, boundary, and exterior directions. Boundary and exterior first derivatives should vanish by CRT orthogonality when $h$, the reciprocal density $1/p$, and the active prefix all factor through $R_d$; inside-face directions reduce to the smaller row.

This target gives stability of proper faces, not decay at the full top face. For $d = P_S$, there is no ambient face protection. The projective closure therefore still requires either

$$\mu^\star_S(\mathcal{H}_S)\,\mathrm{polylog}(P_S) \to 0$$

on the full-conductor/top face, or the hard packet-preserving constraint at every stage.

**Theorem 5.1.3 (Projective closure of variational recovery currents).**
Let

$$J_n(f) = \mathcal{R}_n^{\mathrm{var}} f$$

be the variational recovery currents of Definition 5.1. Assume:

1. the primorial restriction maps

   $$\pi_{m,n} : \mathcal{J}_m \to \mathcal{J}_n, \qquad m \ge n,$$

   are defined on the quotient-character adjacency algebras after quotienting the flat leakage degeneracies;
2. the recovery maps are projectively compatible in the sense that

   $$\pi_{m,n}J_m(f) - J_n(f)$$

   lies in the flat degeneracy space up to an error tending to 0 for each fixed $n$;

3. the leakage forms are lower semicontinuous and compatible under restriction:

$$\Lambda_n(\pi_{m,n} J_m) \le \Lambda_m(J_m) + o_{m,n}(1)$$

for each fixed $n$, with $o_{m,n}(1) \to 0$ as $m \to \infty$;

4. the frame-price scaling condition holds:

$$\mu_n^\star(\mathcal{H}_n)\,\mathrm{polylog}(P_n) \to 0,$$

or the hard packet-preserving constraint $Q_n^\star \in O_{\mathrm{pkt}}$ is imposed at every stage.

Then $J_n(f)$ has a projective limit

$$J_\infty(f) \in \varprojlim_n \mathcal{J}_n/\mathrm{Flat}_n,$$

and this limit is closed:

$$\Lambda_\infty(J_\infty(f)) = 0, \qquad J_\infty(f) \in \mathrm{Crit}_\infty.$$

Equivalently,

$$\Lambda_n(\mathcal{R}_n^{\mathrm{var}} f) \to 0,$$

and the limiting current lies on the projective critical leaf.

*Proof.* Theorem 5.1.2 gives

$$\Lambda_n(J_n(f)) \le C_f\, \mu_n^\star(\mathcal{H}_n)\,\mathrm{polylog}(P_n)$$

in the first-order-priced stratum, and gives exact vanishing in the hard packet-preserving stratum. Under assumption (4), therefore,

$$\Lambda_n(J_n(f)) \to 0.$$

Projective compatibility gives a well-defined element of the inverse limit after quotienting flat degeneracies. For each fixed finite stage $r$, compatibility of the leakage forms gives

$$\Lambda_r(\pi_{n,r} J_n(f)) \le \Lambda_n(J_n(f)) + o_{n,r}(1).$$

Taking $n \to \infty$ yields

$$\Lambda_r(\pi_{\infty,r} J_\infty(f)) = 0$$

in the projective-limit representative. Since this holds at every finite stage, the limiting leakage

$$\Lambda_\infty(J_\infty) := \sup_r \Lambda_r(\pi_{\infty,r} J_\infty)$$

vanishes. Thus $J_\infty(f)$ lies in the closed-current sector $\mathrm{Crit}_\infty$. □

**Conjecture 5.2 (Primorial Noether-Weil Identity).**
There exists a dense test-function class $\mathcal{S}_0$, a cutoff sequence $M_n \to \infty$, and the variational recovery maps of Definition 5.1 (written $\mathcal{R}_n = \mathcal{R}_n^{\mathrm{var}}$),

$$\mathcal{R}_n : \mathcal{S}_0 \to \mathrm{Crit}_n$$

such that, for every $f \in \mathcal{S}_0$,

$$\lim_{n\to\infty} \mathcal{Q}_n(\mathcal{R}_n f) = \mathcal{W}_\zeta(f * f^*).$$

Under this recovery map:

1. additive-character convolution on $R_n$ gives the Dirichlet-sum side;
2. CRT tensor product over the local prime axes gives the Euler-product side;
3. the Noether energy gives the archimedean/spectral positivity;
4. leakage excludes off-critical closed currents.

This is the whole game. If Conjecture 5.2 is proved, the Riemann Hypothesis follows.

**Theorem 5.2.1 (Noether energy identification modulo the finite-place gaps).** Let $J_n(f) = \mathcal{R}_n^{\mathrm{var}} f$, and assume the projective closure hypotheses of Theorem 5.1.3, so that $J_n(f) \to J_\infty(f) \in \mathrm{Crit}_\infty$. Normalize the all-order Noether energy by the Hayes-Mobius coefficient normalization of Definition 5.1 and by the prime-weight lattice measure

$$\alpha = (\alpha_p)_{p \le p_n} \in \mathbb{N}^{\{p \le p_n\}}, \qquad |\alpha|_{\log} = \sum_{p \le p_n} \alpha_p \log p.$$

Assume the two finite-place analytic inputs:

1. **Double cutoff.** The cumulant/occupation cutoff $K_n \to \infty$ and the primorial axis cutoff $P_n \to \infty$ may be chosen so that the squarefree recovery datum of Definition 5.1 extends, through the all-order Hayes-Mobius tower, to the weight lattice $\mathbb{N}^{\{p \le p_n\}}$, and the limits $K_n \to \infty$, $M_n \to \infty$, and $n \to \infty$ commute in the Weil test-function topology.
2. **Primorial uniformity.** The needed prime-number-theorem input in primorial residue classes is uniform enough to pass from the finite $P_n$-smooth recovery sums to the completed von Mangoldt distribution.

Then the finite-place part of the accumulated Noether energy satisfies

$$\lim_{n \to \infty} \mathcal{E}^{(n)}_{\infty,\mathrm{fin}}(J_n(f)) = \mathcal{W}_{\zeta,\mathrm{fin}}(f * f^*),$$

where $\mathcal{W}_{\zeta,\mathrm{fin}}$ denotes the non-archimedean prime/Euler side of Weil's explicit-formula quadratic form. If, in addition, the archimedean lift axioms of Theorem 5.2.2 hold, then

$$\lim_{n \to \infty} \mathcal{E}^{(n)}_{\infty}(J_n(f)) = \mathcal{W}_{\zeta}(f * f^*).$$

*Proof.* By Definition 5.1 and Proposition 5.1.1, $J_n(f)$ is generated from the finite Fourier datum

$$\hat{h}^f_n(\eta) = \sum_{\eta_m = \eta} \tilde{f}\left(\frac{2\pi i \log m}{\log P_n}\right)$$

in the conductor chart selected by the minimizing cascade. The order-$k$ Noether coefficient is the Hayes-Mobius vector

$$\mathsf{C}^{(n)}_k(\hat{h}^f_n) = \sum_{\mathfrak{e},\mathfrak{f}} \sum_{\eta} (\hat{h}^f_n)^{*k}(\eta)\, A^{\mathfrak{f},\mathfrak{e}}_\eta.$$

Thus the all-order Noether energy is a weighted sum of convolution powers of the same finite character datum. The CRT tensor decomposition rewrites those convolution powers by prime axes. After Hayes-Mobius compression, the closed-current projection removes the off-critical mixed leakage terms and keeps precisely the logarithmic Euler-product contribution indexed by the prime-weight lattice:

$$\sum_{\alpha \in \mathbb{N}^{\{p \le p_n\}}} c_\alpha\, \tilde{f}\left(\frac{2\pi i |\alpha|_{\log}}{\log P_n}\right),$$

with $c_\alpha = \Lambda(p^r) = \log p$ on single-axis occupations $\alpha = r\,e_p$ and with the mixed coefficients cancelling on $\mathrm{Crit}_n$. This is the finite Noether analogue of the logarithmic Euler product

$$-\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{r\geq 1} (\log p)\, p^{-rs}.$$

The double-cutoff hypothesis is exactly the assertion that the all-order occupation lift supplies all prime powers $p^r$, rather than only the squarefree axis skeleton visible at finite cumulant order. The primorial uniformity hypothesis is exactly the assertion that the finite sums over $P_n$-smooth/primorial residue data converge, after Mellin inversion, to the von Mangoldt distribution in the Weil explicit formula:

$$\sum_{p,r\geq 1} \log p\, (\hat{f}(r\log p) + \hat{f}(-r\log p)).$$

Therefore the finite-place limit of the Noether energy is $\mathcal{W}_{\zeta,\mathrm{fin}}(f * f^*)$.

It remains only to add the archimedean term. By Theorem 5.1.3 the limiting current is closed. Theorem 5.2.2 then identifies the minimal nonabelian completion of the closed Cartan sector and computes its oscillator-normalized character:

$$\Gamma(s/2)\Gamma((s+1)/2).$$

After inserting the standard powers of $\pi$ and 2 fixed by the completed-zeta normalization, this is exactly the gamma contribution to Weil's form. Adding the finite-place contribution proves the displayed identity with $\mathcal{W}_\zeta(f * f^*)$. □

**Theorem 5.2.2 (Archimedean gauge-lift criterion).**
Let $J_\infty(f) \in \mathrm{Crit}_\infty$ be the closed projective current of Theorem 5.1.3. Suppose the limiting closed-current sector satisfies the following archimedean lift axioms:

1. **Cartan floor.** The finite closed sector contains the two Cartan generators already visible on the cubic row $R = \mathbb{Z}/6\mathbb{Z}$:
$$\mathfrak{h}_\infty \simeq \mathfrak{u}(1)^2.$$
2. **Noether charge is not abelian.** The limiting cubic Noether charge has a nonzero symmetric $d$-symbol component, and this component is preserved under the projective restriction maps.
3. **Closed-current compatibility.** No residual leakage class survives in the projective limit; equivalently, all off-Cartan components that remain must close under the Noether bracket rather than appear as a new carry defect.
4. **Real-cone and Fisher-volume preservation.** The archimedean completion acts by positive-cone-preserving transformations that preserve the limiting Fisher volume form.
5. **Critical involution.** The representation carries an involution
$$\iota : s \longleftrightarrow 1 - s$$
which exchanges the two closed Cartan branches.
6. **Minimal positive-energy completion.** Among nonabelian completions satisfying items 1–5, the projective lift is the minimal semisimple positive-energy completion.

Then the Cartan floor is forced to complete to

$$\mathfrak{g}_\infty^{\min} \simeq \mathfrak{su}(2) \oplus \mathfrak{su}(3)$$

up to compact real form and harmless central/flat factors. Moreover, under the oscillator normalization in which the two closed Cartan branches have Mellin exponents $s/2$ and $(s+1)/2$, the archimedean character of the lift contributes

$$\Gamma(s/2)\Gamma((s+1)/2)$$

to the completed Noether-Weil form, up to the standard powers of $\pi$ and 2 fixed by the chosen completed-zeta normalization.

*Proof.* The finite cubic row gives two closed Cartan generators, hence the abelian floor $\mathfrak{h}_\infty \simeq \mathfrak{u}(1)^2$. If the projective limit remained purely abelian, every Noether bracket on the surviving sector would be diagonal in the Cartan weights and the symmetric cubic $d$-symbol would be either zero or an external tensor not generated by the current algebra. This contradicts item 2 together with item 3: a nonzero cubic charge that is not closed by the algebra would be precisely residual leakage, while Theorem 5.1.3 has placed the limit in the closed sector. Therefore the archimedean completion must add root directions.

Real-cone and Fisher-volume preservation rule out arbitrary noncompact or volume-distorting extensions of the Cartan floor. The smallest compact semisimple extension carrying a nontrivial rank-one reflection that can swap the two branches is the $\mathfrak{su}(2)$ factor. The smallest compact semisimple extension carrying a nonzero invariant symmetric cubic tensor is $\mathfrak{su}(3)$: the $\mathfrak{su}(2)$ adjoint has no independent rank-three symmetric invariant, while $\mathfrak{su}(3)$ has the standard $d_{abc}$ tensor. Minimality in item 6 therefore forces

$$\mathfrak{g}_\infty^{\min} \simeq \mathfrak{su}(2) \oplus \mathfrak{su}(3)$$

after quotienting central and flat factors invisible to the leakage form. The $\mathfrak{su}(2)$ factor supplies the branch-exchange involution $s \leftrightarrow 1-s$; the $\mathfrak{su}(3)$ factor supplies the cubic Noether charge compatible with the closed current.

It remains to compute the archimedean character. In the oscillator normalization, the two branch radial variables $x, y > 0$ have character density

$$\Theta_s(x,y) = e^{-x-y}\, x^{s/2-1}\, y^{(s+1)/2-1}.$$

This is the unique positive-energy density compatible with the two Cartan weights, Fisher-volume preservation, and the critical involution exchanging the two branches. Mellin integration gives

$$\int_0^\infty \int_0^\infty e^{-x-y} x^{s/2-1} y^{(s+1)/2-1}\, dx\, dy = \Gamma(s/2)\Gamma((s+1)/2).$$

The omitted factors are exactly the conventional powers of $\pi$ and 2 introduced when passing from this oscillator normalization to the standard completed zeta normalization; the duplication formula fixes their placement. Thus the closed-current nonabelian lift contributes precisely the gamma factor required in the completed Weil form. □

**Target Package 5.2.3 (Bures/projective verification of the Archimedean lift axioms).** Theorem 5.2.2 is a conditional criterion. To turn it into the Archimedean Noether Limit, the following statements must be proved from the projective finite tower together with the Bures completion of Paper 18:

1. **Cartan floor from the cubic row.** The closed-current sector on $R = \mathbb{Z}/6\mathbb{Z}$ embeds projectively as exactly two classical abelian branches:

$$\mathfrak{h}_{\text{floor}} \simeq \mathfrak{u}(1)^2.$$

   Paper 17 supplies the exhaustion of real-simplex cubic charges; the required pass condition is that no additional real-cone-preserving classical cubic charge survives the projective restrictions.
2. **Internal cubic $d$-symbol.** The limiting Amari-Chentsov/Noether cubic charge must be identified with the invariant symmetric tensor of the Bures-unitary $\mathfrak{su}(3)$ sector and must survive all projective restrictions. It is not enough to attach a cubic tensor externally: the $d_{abc}$ component must be generated by the closed current algebra. Proposition 5.2.3a and Corollary 5.2.3b construct the compatible Bures Cartan-floor maps and prove the Bures-side cubic scalar is nonzero; Proposition 5.2.4a proves the finite $R = 6$ $A_2$-matching calculation.
3. **No residual leakage upstairs.** Theorem 5.1.3 gives $\Lambda_\infty(J_\infty) = 0$ on the diagonal shadow. The Bures lift must show that every off-Cartan component surviving upstairs closes under the Noether bracket, while every nonclosed component is killed by critical-leaf projection. Thus no carry or leakage class may remain in the inverse limit.
4. **Real cone and Fisher volume.** Paper 18 shows that the Bures-Helstrom metric restricts to Fisher on diagonal density matrices and that local unitaries preserve trace, positivity, the Bures metric, and CRT tensor structure. The required pass condition is exact recovery of the diagonal simplex, Fisher volume as the diagonal shadow of Bures volume, and no noncompact or volume-distorting archimedean extension.
5. **Critical involution.** The involution $s \leftrightarrow 1-s$ must be realized as an actual Weyl/reflection or modular-conjugation symmetry of the Bures positive-energy representation, exchanging the two closed Cartan branches while preserving the completed-zeta normalization.
6. **Minimal positive-energy completion.** One must prove that no smaller semisimple positive-energy algebra satisfies branch exchange, nonzero internal cubic $d$-symbol, real-cone preservation, and Fisher/Bures volume preservation. Any quotient taken in the proof may remove only genuinely central or leakage-flat factors.
7. **Representation-theoretic character.** The formal Mellin integral in the proof of Theorem 5.2.2 must be upgraded to a character calculation: the two radial variables $x, y > 0$ must be identified with the lifted Cartan branches, the exponents $s/2$ and $(s+1)/2$ must be derived from the positive-energy weight or half-density normalization, and the conventional powers of 2 and $\pi$ must be accounted for by the completed-zeta normalization and the duplication formula. Proposition 5.2.5a proves the even/odd oscillator radial measures, and Theorem 5.2.5b gives the metaplectic realization once the projective Bures limit is shown to select the Weyl/CCR positive-energy representation. Proposition 5.2.5c proves that this selection is not automatic from the current projective Bures data; Selection Axiom 5.2.5d states the additional representation input needed by the oscillator route.

**Rank watchpoint.**
$\mathfrak{su}(2) \oplus \mathfrak{su}(3)$ has rank 3. Therefore $\mathfrak{u}(1)^2$ should be read as a two-dimensional Cartan floor inside the completed Cartan data, not as the full maximal torus of the completed semisimple algebra. The Archimedean Noether Limit must either identify the extra Cartan direction and show that it is central, leakage-flat, or normalization-invisible, or explicitly include its positive-energy role in the character calculation. Until this rank bookkeeping is proved, the gauge-lift route remains conditional.

**Proposition 5.2.3a (canonical Bures Cartan-floor lift).**

Assume the projective restriction maps of Definition 5.1.3a and the closed-sector compatibility of Theorem 5.1.3. Let $R_2 = \mathbb{Z}/6\mathbb{Z}$ be the cubic row, and let

$$\mathrm{Cart}_6 \subset \mathrm{Crit}_2/\mathrm{Flat}_2$$

be the two-dimensional Cartan floor supplied by Paper 17, Corollary 2.4. Fix the standard Cartan basis $T_3, T_8$ of $\mathfrak{su}(3)$ with

$$T_a = \frac{\lambda_a}{2}, \qquad \mathrm{Tr}_3(T_a T_b) = \frac{1}{2}\delta_{ab}.$$

Choose the unique scale $s_B > 0$ for which the map

$$\iota_{A_2} : \mathrm{Cart}_6 \longrightarrow \mathfrak{Herm}_0(\mathbb{C}^2 \otimes \mathbb{C}^3), \qquad c_3 \mapsto s_B\, I_2 \otimes T_3, \quad c_8 \mapsto s_B\, I_2 \otimes T_8$$

pulls the Bures metric at $\rho_0 = I_6/6$ back to the chosen Fisher normalization on $\mathrm{Cart}_6$. For $n \geq 2$, define

$$\jmath_n^{\mathrm{floor}} = \iota_{A_2} \circ \pi_{n,2}^{\mathrm{Cart}},$$

where $\pi_{n,2}^{\mathrm{Cart}}$ is primorial restriction to the $R = 6$ Cartan floor.

Then the maps $\jmath_n^{\mathrm{floor}}$ are projectively compatible:

$$\jmath_m^{\mathrm{floor}} = \jmath_n^{\mathrm{floor}} \circ \pi_{m,n}$$

after restriction to the Cartan-floor quotient, for $m \geq n \geq 2$. Their image carries a nonzero Bures $\mathfrak{su}(3)$ symmetric cubic:

$$d_{338} = \frac{1}{\sqrt{3}}, \qquad d_{888} = -\frac{1}{\sqrt{3}}$$

in the unscaled Gell-Mann convention, and the scale $s_B$ only multiplies this cubic by $s_B^3 \neq 0$.

*Proof.* Functoriality of $\pi$ gives

$$\pi_{m,2}^{\mathrm{Cart}} = \pi_{n,2}^{\mathrm{Cart}} \pi_{m,n}$$

on the Cartan-floor quotient, hence the displayed compatibility of $\jmath_n^{\mathrm{floor}}$. At $\rho_0 = I_6/6$,

$$g^B_{\rho_0}(A, B) = 6\,\mathrm{Tr}_6(AB).$$

For $A = I_2 \otimes X$, $B = I_2 \otimes Y$, this is

$$g^B_{\rho_0}(I_2 \otimes X, I_2 \otimes Y) = 12\,\mathrm{Tr}_3(XY),$$

so a unique positive scale $s_B$ matches any fixed Fisher normalization on the two-dimensional floor. The standard $\mathfrak{su}(3)$ Jordan identity then gives the nonzero $d$-symbol values on $T_3, T_8$. □

This proposition proves the existence of a compatible Bures lift of the Cartan floor and a nonzero Bures cubic on its image. It does not by itself prove that the finite Noether cubic equals that Bures cubic; that is the separate $A_2$-matching problem below.

**Corollary 5.2.3b (nonzero Bures cubic scalar on the floor).**
With $\jmath_n^{\mathrm{floor}}$ as in Proposition 5.2.3a, define

$$\mathcal{D}^{B,\mathrm{floor}}_{\rho_0}(X, Y, Z) = D_{\mathfrak{su}(3)}(s_B^{-1}\,\mathrm{pr}_{A_2} X,\, s_B^{-1}\,\mathrm{pr}_{A_2} Y,\, s_B^{-1}\,\mathrm{pr}_{A_2} Z)$$

on the image of the Cartan-floor lift, where $\mathrm{pr}_{A_2}$ removes the trivial $I_2$-factor. Equivalently,

$$\mathcal{D}^B_{\rho_0}(s_B I_2 \otimes X, s_B I_2 \otimes Y, s_B I_2 \otimes Z) = c_B^{\mathrm{Bures}} D_{\mathfrak{su}(3)}(X, Y, Z), \qquad c_B^{\mathrm{Bures}} = s_B^3 \neq 0$$

if the unscaled finite Cartan coordinates are used. Thus the Bures lift itself carries a nonzero internal $A_2$ cubic scalar.

*Proof.* Proposition 5.2.3a gives $s_B > 0$ and identifies the lifted floor with $s_B(I_2 \otimes \mathrm{span}\{T_3, T_8\})$. The trace-free Jordan cubic is homogeneous of degree three, hence changing from unscaled finite Cartan coordinates to Bures coordinates multiplies it by $s_B^3$. Since $D_{\mathfrak{su}(3)}(T_3, T_3, T_8) = 1/\sqrt{3}$, this scalar is nonzero. □

This proves $c_B \neq 0$ on the Bures side. The remaining issue is not the existence of a nonzero Bures cubic; it is whether the finite closed Noether cubic $Q_6$ is exactly the pullback of this internal Bures-Jordan cubic. That finite-to-Bures identification is proved for the $R = 6$ $A_2$ floor in Proposition 5.2.4a.

**Candidate Theorem 5.2.4 (internal $d$-symbol survival via $A_2$ matching).**
Let

$$\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^3, \qquad \rho_0 = \frac{I_{\mathcal{H}}}{6},$$

and choose the Gell-Mann normalization

$$\mathrm{Tr}_3(\lambda_a \lambda_b) = 2\delta_{ab}, \qquad T_a = \frac{\lambda_a}{2} \in \mathfrak{su}(3).$$

On the $\mathfrak{su}(3)$ factor define the trace-free Jordan product

$$X \circ_0 Y = \frac{1}{2}(XY + YX) - \frac{1}{3}\,\mathrm{Tr}_3(XY) I_3$$

and the symmetric cubic tensor

$$D_{\mathfrak{su}(3)}(X, Y, Z) = 4\,\mathrm{Tr}_3((X \circ_0 Y)Z).$$

In the basis $T_a$, this is the standard invariant tensor

$$D_{\mathfrak{su}(3)}(T_a, T_b, T_c) = d_{abc},$$

up to the fixed normalization convention. In particular it is nonzero on the Cartan of $\mathfrak{su}(3)$; for the usual basis,

$$d_{338} = \frac{1}{\sqrt{3}}, \qquad d_{888} = -\frac{1}{\sqrt{3}}.$$

Assume the Cartan-floor lift $\jmath_n^{\mathrm{floor}}$ of Proposition 5.2.3a extends to the closed archimedean sector and satisfies the $A_2$-matching condition:

1. $\jmath_n^{\mathrm{floor}}$ intertwines primorial restriction with the Bures diagonal measurement channel;
2. the pullback of the Bures metric at $\rho_0$ is the limiting Fisher metric on the closed current sector;
3. the normalized cubic Noether tensor on $\mathrm{Crit}_n$ converges, after pullback by $\jmath_n^{\mathrm{floor}}$, to the Bures-Jordan tensor
$$\mathcal{D}^B_{\rho_0}(I_2 \otimes X, I_2 \otimes Y, I_2 \otimes Z) = c_B\, D_{\mathfrak{su}(3)}(X, Y, Z)$$
with $c_B \neq 0$;

4. the convergence is functorial under the restriction maps $\pi_{m,n}$.

Then the limiting cubic Noether charge has a nonzero internal $\mathfrak{su}(3)$ $d$-symbol component, and this component is preserved under projective restriction. It is internal because it is generated by the Jordan product in the Bures-unitary algebra, not appended as an external cubic tensor.

*Proof sketch.* The identity

$$\{T_a, T_b\} = \frac{1}{3}\delta_{ab} I_3 + d_{abc} T_c$$

is the standard $\mathfrak{su}(3)$ Jordan decomposition. After removing the scalar part, the only remaining symmetric invariant cubic on the adjoint is the $d$-tensor. Assumptions (1)–(3) identify the normalized closed Amari-Chentsov/Noether cubic with this trace-free Jordan cubic up to the nonzero scalar $c_B$. Assumption (4) gives projective survival. Thus the $d$-symbol is generated by the closed Bures current algebra itself. □

**Failure mode.** Proposition 5.2.4a rules out $c_B = 0$ on the finite $R = 6$ $A_2$ floor, and Proposition 5.2.4b below rules out disappearance of the corresponding cylinder class in the projective tower. The remaining failure modes are representation-theoretic: the full closed Bures sector may fail to select the Weyl/metaplectic oscillator representation, or an off-floor closed sector may converge to a tensor not generated by the Bures Jordan product. In either case Theorem 5.2.2 cannot yet be invoked.

**Proposition 5.2.4a (the $R = 6$ $A_2$ cubic matching).**
Let $Q_6$ be the normalized closed cubic Noether form on the $R = 6$ Cartan floor, with the Paper 17 Amari-Chentsov sign convention denoted by $\varepsilon_{CH} \in \{\pm 1\}$. At the centroid

$$p_* = \frac{1}{6}\mathbf{1}$$

the categorical cubic is

$$T^{CH}_{p_*}(a, b, c) = 36 \sum_{r \in R} a_r b_r c_r$$

up to the same global sign $\varepsilon_{CH}$. Identify the $A_2$ Cartan floor by

$$X \in \operatorname{span}\{T_3, T_8\} \longmapsto h_X = s_B \operatorname{diag}(I_2 \otimes X) \in T_{p_*}\Delta_R,$$

where $s_B$ is the Bures/Fisher scale of Proposition 5.2.3a. Then, for all $X, Y, Z \in \operatorname{span}\{T_3, T_8\}$,

$$Q_6(h_X, h_Y, h_Z) = 18\, \varepsilon_{CH} s_B^3\, D_{\mathfrak{su}(3)}(X, Y, Z).$$

Equivalently, after transporting the finite coordinates back to the unscaled $A_2$ Cartan,

$$Q_6(X, Y, Z) = c_B D_{A_2}(X, Y, Z), \qquad c_B = 18\, \varepsilon_{CH} s_B^3 \neq 0.$$

With the positive Amari-Chentsov convention, $c_B = 18 s_B^3$. If the definition of $Q_6$ absorbs the autonomous energy factor $\frac{1}{3}T^{CH}$ from Paper 17, Definition 2.1, the same matching holds with $c_B = 6\, \varepsilon_{CH} s_B^3$; only the global scalar convention changes.

In the diagonal coordinate $X = xT_3 + yT_8$, this becomes

$$Q_6(X, X, X) = \frac{18\, \varepsilon_{CH} s_B^3}{\sqrt{3}}(3x^2 y - y^3),$$

since

$$D_{\mathfrak{su}(3)}(X,X,X) = \frac{1}{\sqrt{3}}(3x^2y - y^3).$$

*Proof.* On the probability simplex, the Amari-Chentsov tensor at a point $p$ is

$$T_p^{CH}(a,b,c) = \sum_{r\in R} \frac{a_r b_r c_r}{p_r^2}$$

up to the fixed global sign convention. At $p_* = (1/6)\mathbf{1}$, this is

$$36 \sum_{r\in R} a_r b_r c_r.$$

For $h_X = s_B \operatorname{diag}(I_2 \otimes X)$, and similarly for $Y, Z$, the six diagonal entries are two identical copies of the three diagonal entries of $s_B X$. Therefore

$$\sum_{r\in R} (h_X)_r (h_Y)_r (h_Z)_r = 2 s_B^3 \operatorname{Tr}_3(XYZ).$$

Hence

$$Q_6(h_X, h_Y, h_Z) = \varepsilon_{CH}\, 36 \cdot 2 s_B^3 \operatorname{Tr}_3(XYZ) = 72 \varepsilon_{CH} s_B^3 \operatorname{Tr}_3(XYZ).$$

On the traceless diagonal Cartan of $\mathfrak{su}(3)$,

$$X \circ_0 Y = XY - \frac{1}{3} \operatorname{Tr}_3(XY) I_3,$$

and $\operatorname{Tr}_3 Z = 0$, so

$$D_{\mathfrak{su}(3)}(X,Y,Z) = 4 \operatorname{Tr}_3((X \circ_0 Y)Z) = 4 \operatorname{Tr}_3(XYZ).$$

Thus

$$Q_6(h_X, h_Y, h_Z) = 18 \varepsilon_{CH} s_B^3 D_{\mathfrak{su}(3)}(X,Y,Z),$$

and the scalar is nonzero because $s_B > 0$. Substituting

$$T_3 = \operatorname{diag}(1/2, -1/2, 0), \qquad T_8 = \operatorname{diag}(1/(2\sqrt{3}), 1/(2\sqrt{3}), -1/\sqrt{3})$$

gives the displayed binary cubic. □

This solves the finite $A_2$-matching calculation. The remaining part of Candidate Theorem 5.2.4 is projective, not algebraic: extend the floor matching functorially to the whole closed Archimedean sector and prove that all non-floor closed components are also generated internally by the Bures Jordan algebra.

**Proposition 5.2.4b (projective survival of the matched $A_2$ class).**
Let $S_2 = \{2,3\}$, let $S_n = \{p_1, \dots, p_n\}$, and let

$$q_2 := Q_6 \in \operatorname{Sym}^3(\operatorname{Cart}_6^*) \subset \mathcal{J}_{S_2}/\operatorname{Flat}_{S_2}$$

be the $A_2$-matched cubic of Proposition 5.2.4a. For $n \geq 2$, define its cylinder lift by

$$q_n := (\iota_{S_2,S_n})_* q_2,$$

meaning that every character/tensor leg is extended by the normalized CRT pullback of Definition 5.1.3a and the result is read in $\mathcal{J}_{S_n}/\mathrm{Flat}_{S_n}$. Then

$$\pi_{S_m,S_n} q_m = q_n, \qquad m \geq n \geq 2,$$

and therefore $q_\infty = (q_n)_{n\geq 2}$ is a well-defined nonzero element of the projective closed sector:

$$q_\infty \in \varprojlim_{n\geq 2} \left(\mathrm{Crit}_{S_n}/\mathrm{Flat}_{S_n}\right), \qquad \pi_{\infty,S_2} q_\infty = q_2 \neq 0.$$

Moreover, under the Bures Cartan-floor maps,

$$j_n^{\mathrm{floor}}(q_n) = c_B D_{A_2}, \qquad c_B = 18\,\varepsilon_{CH} s_B^3$$

up to the same global energy-normalization convention as in Proposition 5.2.4a.

*Proof.* In normalized CRT character bases, Definition 5.1.3a gives

$$r_{S,T} e_\eta^S = \begin{cases} e_{\eta|T}^T, & \eta_{S\setminus T} = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Thus for $S_2 \subset S_n \subset S_m$,

$$r_{S_m,S_n} \iota_{S_2,S_m} = \iota_{S_2,S_n}$$

on every tensor leg. Applying this identity to the three legs of $q_2$ gives $\pi_{S_m,S_n} q_m = q_n$. Hence the family $(q_n)$ is a projective system. Its $S_2$-coordinate is $q_2$, which is nonzero by Proposition 5.2.4a, so the inverse-limit class is nonzero. Closed-sector stability under restriction is exactly the closed-sector compatibility hypothesis already used in Proposition 5.2.3a and Criterion 5.1.3b. Finally, $j_n^{\mathrm{floor}} = \iota_{A_2} \pi_{n,2}^{\mathrm{Cart}}$, so each cylinder lift is sent back to the same $A_2$ Bures cubic computed in Proposition 5.2.4a. □

This proves survival of the matched $d$-symbol class in the projective tower. What it does not prove is that the whole projective Bures representation is the metaplectic oscillator representation; that is the separate representation-selection problem below.

**Candidate Theorem 5.2.5 (oscillator character for the gamma factor).**
Assume the two closed Cartan branches of Target Package 5.2.3 act in the positive-energy oscillator model as the even ground branch and the odd first excited branch. In Gaussian normalization, take radial representatives

$$\psi_0(x) = e^{-\pi x^2}, \qquad \psi_1(y) = y\, e^{-\pi y^2}, \qquad x, y > 0.$$

Their Mellin character is

$$\chi_{\mathrm{osc}}(s) = 4 \int_0^\infty \int_0^\infty e^{-\pi(x^2+y^2)} x^{s-1} y^s \, dx\, dy.$$

Equivalently, after the change of variables $X = \pi x^2$, $Y = \pi y^2$, this has oscillator-normalized density

$$e^{-X-Y} X^{s/2-1} Y^{(s+1)/2-1}\, dX\, dY.$$

Therefore

$$\chi_{\mathrm{osc}}(s) = \pi^{-(2s+1)/2} \Gamma(s/2) \Gamma((s+1)/2),$$

up to the harmless scalar convention chosen in the half-line normalization. Removing the Gaussian powers of $\pi$, or equivalently passing to the oscillator normalization used in Theorem 5.2.2, gives

$$\Gamma(s/2)\Gamma((s+1)/2).$$

The duplication formula

$$\Gamma(s/2)\Gamma((s+1)/2) = 2^{1-s}\sqrt{\pi}\,\Gamma(s)$$

then fixes the placement of the conventional powers of 2 and $\pi$ when matching the completed-zeta normalization.

The critical involution $s \leftrightarrow 1-s$ is not a formal symmetry of the integral alone. It must be implemented by the oscillator Fourier/Weyl operator, or by an equivalent modular conjugation, exchanging the two closed Cartan branches and preserving the completed normalization. Once this representation symmetry is supplied, the Mellin transform above is a genuine positive-energy character calculation rather than an inserted gamma factor.

**Proposition 5.2.5a (even/odd oscillator radial measures).**
In the one-dimensional positive-energy oscillator representation, let the even and odd branch rays be represented on the half-line by

$$\psi_0(x) = e^{-\pi x^2}, \qquad \psi_1(y) = ye^{-\pi y^2}, \qquad x, y > 0.$$

Then their Mellin radial characters are

$$\chi_+(s) = 2\int_0^\infty e^{-\pi x^2} x^{s-1}\,dx = \pi^{-s/2}\Gamma(s/2),$$

and

$$\chi_-(s) = 2\int_0^\infty e^{-\pi y^2} y^{s}\,dy = \pi^{-(s+1)/2}\Gamma((s+1)/2).$$

Consequently the two-branch product is

$$\chi_+(s)\chi_-(s) = \pi^{-(2s+1)/2}\Gamma(s/2)\Gamma((s+1)/2).$$

After the changes of variables $X = \pi x^2$ and $Y = \pi y^2$, the two Mellin measures become

$$e^{-X}X^{s/2-1}\,dX, \qquad e^{-Y}Y^{(s+1)/2-1}\,dY$$

up to the displayed powers of $\pi$. In oscillator normalization this is exactly the density

$$e^{-X-Y}X^{s/2-1}Y^{(s+1)/2-1}\,dX\,dY.$$

*Proof.* The first formula is the standard Gaussian Mellin integral

$$\int_0^\infty e^{-\pi x^2} x^{s-1}\,dx = \frac{1}{2}\pi^{-s/2}\Gamma(s/2),$$

with the factor 2 coming from passage from the even full-line ray to the positive radial half-line. For the odd branch, the first Hermite ray has one radial power $y$, so the Mellin weight is $y^{s-1}\cdot y = y^s$, and

$$\int_0^\infty e^{-\pi y^2} y^{s}\,dy = \frac{1}{2}\pi^{-(s+1)/2}\Gamma((s+1)/2).$$

Multiplication gives the product character. The substitutions $X = \pi x^2$, $Y = \pi y^2$ give the two radial measures stated above. □

**Theorem 5.2.5b (metaplectic realization criterion).**
Let $\mathcal{B}_\infty^{A_2}$ be the Hilbert completion of the surviving closed $A_2$ floor generated by Proposition 5.2.4b under the limiting Bures metric. Suppose the projective Bures limit supplies:

1. a strongly continuous Weyl representation

$$W(u, v), \qquad (u, v) \in \mathbb{R}^2,$$

   whose infinitesimal position and momentum branches are the two closed Cartan-floor branches;
2. the canonical commutation relation

$$W(u, v)W(u', v') = e^{\pi i(uv' - vu')}W(u + u', v + v');$$

3. a positive-energy number operator $N$ with cyclic ground ray $\psi_0$ and first odd ray $\psi_1$, such that the Bures diagonal measurement channel sends these rays to the half-line radial densities of Proposition 5.2.5a;
4. the critical branch involution is implemented by the metaplectic Fourier operator

$$\mathcal{F}W(u, v)\mathcal{F}^{-1} = W(v, -u);$$

5. the extra rank-one Cartan direction in $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$ is central, leakage-flat, normalization-invisible, or else contributes a separately accounted positive-energy factor.

Then $\mathcal{B}_\infty^{A_2}$ is unitarily equivalent, on the cyclic closed sector, to the Schrodinger oscillator representation. Under this equivalence the two Cartan branches are realized by the even ground ray and odd first Hermite ray, and the Archimedean character is

$$\pi^{-(2s+1)/2}\Gamma(s/2)\Gamma((s + 1)/2),$$

or $\Gamma(s/2)\Gamma((s + 1)/2)$ in oscillator normalization.

*Proof.* Assumptions (1)–(2) give a regular irreducible Weyl system on the cyclic closed sector generated by the Cartan-floor branches. By the Stone-von Neumann theorem it is unitarily equivalent to the Schrodinger representation of the Heisenberg group on $L^2(\mathbb{R})$, with $u$ and $v$ acting as the usual position and momentum translations. Assumption (4) identifies the branch involution with the metaplectic Fourier operator, the standard lift of the symplectic rotation $(u, v) \mapsto (v, -u)$. Assumption (3) selects the positive-energy oscillator polarization and its parity splitting: $\psi_0(x) = e^{-\pi x^2}$ is the even ground ray and $\psi_1(x) = xe^{-\pi x^2}$ is the first odd ray. Proposition 5.2.5a then computes the radial Mellin character. Assumption (5) ensures no unaccounted Cartan factor changes the character. □

**Proposition 5.2.5c (no-go for automatic Weyl/CCR selection).**
The projective Bures tower data proved above do not, by themselves, force the Weyl/CCR positive-energy oscillator representation. More precisely, the following data:

1. the normalized CRT restriction maps and projective survival of the $A_2$-matched cylinder class $q_\infty$;
2. the Bures/Fisher metric compatibility on the Cartan floor;

3. the nonzero internal $A_2$ cubic

$$q_\infty = c_B D_{A_2}, \qquad c_B \neq 0;$$

4. leakage extinction on the diagonal projective critical leaf;

do not determine a strongly continuous Weyl system, a nontrivial central CCR cocycle, a positive-energy number operator, or the metaplectic Fourier operator. Therefore the requested Weyl/CCR selection theorem cannot be proved from the current finite/projective Bures axioms alone.

*Proof.* At every finite row $S_n$, the Bures lift acts through a finite matrix algebra of traceless Hermitian operators. If two finite-dimensional operators $P_n, Q_n$ satisfied an exact CCR

$$[Q_n, P_n] = i\hbar I, \qquad \hbar \neq 0,$$

then taking traces would give

$$0 = \operatorname{Tr}[Q_n, P_n] = i\hbar \operatorname{Tr} I,$$

a contradiction. Hence no exact CCR is present at any finite stage.

An exact CCR can only appear after an additional limiting operation that specifies an unbounded generator pair and a nontrivial central two-cocycle. But the projective data listed above are symmetric/metric and cubic: they determine the Bures inner product on the floor and the symmetric $A_2$-Jordan tensor. They do not determine an antisymmetric symplectic form

$$\omega((u, v), (u', v')) = uv' - vu',$$

nor the multiplier

$$e^{\pi i(uv' - vu')}$$

in the Weyl relations.

The nonuniqueness is concrete. Starting from any Hilbert realization of the surviving cylinder class, one may form a direct sum with an additional leakage-flat or shadow-invisible positive Hilbert sector. The finite Cartan-floor projections, Bures metric shadow, and symmetric $A_2$ cubic on the visible sector are unchanged, but the resulting representation is not the irreducible Schrodinger oscillator representation unless one imposes cyclicity/irreducibility and removes the invisible summand. Even on the visible sector, a commuting radial representation can reproduce the same positive half-line densities while carrying no Weyl multiplier at all. Thus the symmetric Bures/Jordan data and projective survival of $q_\infty$ are insufficient to select the Weyl/CCR representation.

Consequently Theorem 5.2.5b is sharp: its Weyl system, CCR cocycle, positive-energy polarization, metaplectic involution, and extra-Cartan normalization hypotheses are genuinely additional representation-selection conditions, not consequences of the current projective Bures tower. □

**Selection Axiom 5.2.5d (metaplectic Archimedean selection).**
To close the Archimedean Noether Limit by the oscillator route, one must add or prove the following representation-selection statement:

The projective Bures completion of the surviving $A_2$ cylinder class admits a regular irreducible Weyl representation with cocycle

$$W(u, v)W(u', v') = e^{\pi i(uv' - vu')} W(u + u', v + v'),$$

the critical branch involution is the metaplectic Fourier lift

$$\mathcal{F}W(u,v)\mathcal{F}^{-1} = W(v,-u),$$

and the Bures positive-energy polarization sends the two closed Cartan branches to the even ground ray and odd first Hermite ray. Under this axiom, Theorem 5.2.5b applies and the Archimedean character is

$$\Gamma(s/2)\Gamma((s+1)/2)$$

in oscillator normalization.

**Failure mode.** If the projective Bures representation does not identify the two closed branches with the even/odd oscillator radial measures, or if the $s \leftrightarrow 1-s$ symmetry is not implemented by an actual operator on the positive-energy representation, then the gamma calculation remains a formal Mellin ansatz rather than an Archimedean Noether Limit.

### 5.3 Low-Order Verification

The identity is exact at the first two nontrivial stages.

**Quadratic order.** On $R_1 = \mathbb{Z}/2\mathbb{Z}$, the quadratic Noether energy is the Fisher energy on the unique nontrivial conductor packet. Its Gram matrix is the centered simplex Gram

$$\langle v_a, v_b\rangle = \delta_{ab} - \frac{1}{2}.$$

By the Fisher-conductor bridge at the centroid, this is the leading quadratic term of the completed explicit formula, i.e. the $L^2(dx/x)$ archimedean main term after completion. The leakage form vanishes on the unique closed current. Therefore

$$\mathcal{Q}_1(\mathcal{R}_1 f) = \mathcal{W}_\zeta(f * f^*) + O(\text{higher cumulants})$$

with exact equality at quadratic truncation.

**Cubic order.** On $R_2 = \mathbb{Z}/6\mathbb{Z}$, the conductor packets are

$$\{2\}, \qquad \{3\}, \qquad \{6\}.$$

The cubic action-energy theorem of Paper 17 gives

$$\mathcal{E}_3 = \tfrac{1}{2}g(v,v) + \tfrac{1}{3}T(h,v,v), \qquad T = K_{CH}.$$

At the centroid this row has the explicit packet certificate

$$A_2 = \{k=3\}, \qquad A_3 = \{k=2,4\}, \qquad M = \{k=1,5\},$$

with real dimensions $1,2,2$, and the Amari-Chentsov cubic has the selection rule $k+\ell+m \equiv 0 \pmod 6$. Pure $A_3$ cubic terms close through $(2,2,2)$ and $(4,4,4)$, pure $A_2$ self-coupling is absent, and $A_2$-$A_3$ transfer requires the mixed conductor-6 packet $M$, via $(2,3,1)$ or $(4,3,5)$. Therefore the projection $\Pi_\tau$ onto $A_2 \oplus A_3$ preserves the single-prime/Cartan time-translation sector and localizes the finite typed residual on $M$. Constraint 3 depth reuse then says that embeddings into $Z/12Z$ and $Z/18Z$ should not open new prime-power-depth residual axes above this $P_2$ floor. The only

admissible closed currents are the two Cartan $\mathfrak{u}(1)$ generators (Paper 17, Corollary 2.4). Expanding the cubic term in additive-character coordinates recovers the first prime correction

$$\sum_{p=2,3} \log p \, (\hat{f}(\log p) + \hat{f}(-\log p))$$

together with the corresponding completed archimedean correction. The mixed $2 \cdot 3$ term is fixed by the Hayes-Mobius vector $\mathsf{C}_3^{(2)}$ and vanishes after closed-leaf projection, as required by the functional-equation side. Thus

$$\mathcal{Q}_2(\mathcal{R}_2 f) = \mathcal{W}_\zeta(f * f^*)\big|_{\leq 3}$$

at cubic truncation. Positivity and boundedness on this finite $P_2$ row should be read as the order-3 instance of the all-order positivity program above, not as an unconditional proof of RH.

**Quartic order.** On $R_3 = \mathbb{Z}/30\mathbb{Z} \cong \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$, the first three-prime row, the fourth cumulant $Q_{ijk\ell} = \partial^4 \Phi$ supplies the order-4 contribution. Read at the centroid in the additive-character basis (the method of §5.3 / T3, one order higher), its selection rule is

$$Q(\chi_a, \chi_b, \chi_c, \chi_d)\big|_u = \delta_{a+b+c+d \equiv 0} - \delta_{a+b \equiv 0}\delta_{c+d \equiv 0} - \delta_{a+c \equiv 0}\delta_{b+d \equiv 0} - \delta_{a+d \equiv 0}\delta_{b+c \equiv 0},$$

the connected quadruple sum-rule minus the three Wick (Gaussian) pairings — verified to machine precision (certificate below). This is the **first cumulant order to carry such a subtraction**: the cubic order above has none, its only proper sub-partition isolating a single character that vanishes on $H_0$. The three Wick terms are exactly the order-2 (Fisher/Gaussian) content already carried by $\mathcal{Q}_1$ at quadratic order; the cumulant subtracts them, leaving the *connected* part — quadruples with $a + b + c + d \equiv 0$ and no vanishing sub-pair — as the genuine quartic content. Exact enumeration on $\mathbb{Z}/30$ gives 1082 connected quadruples against 120 Wick-reducible, of which **914** couple all three primes $\{2, 3, 5\}$ and reduce to neither a Gaussian pairing nor a pairwise (two-prime) face: the first non-pairwise prime interaction in the tower, the obstruction §7 anticipates, now explicit and counted. The order-4 truncation of the Noether form then identifies, through the closed-current projection of Theorem 5.2.1, with

$$\mathcal{Q}_3(\mathcal{R}_3 f) = \mathcal{W}_\zeta(f * f^*)\big|_{\leq 4},$$

and two distinct structures meet here. The prime $p = 5$ enters the recovered von Mangoldt distribution through the **single-axis** Cartan conductor packet $A_5$ (conductor 5, available because $5 \leq p_3$; Hayes-Mobius weight $\Lambda(5) = \sum_{e|5} \mu(5/e) \log e = \log 5$), extending the cubic single-axis recovery $\sum_{p=2,3} \log p$ to $\sum_{p=2,3,5}$. The *connected three-prime* quartic vertices, by contrast, are **mixed** (multi-axis) occupations, on which $\Lambda = 0$; they are the cross-prime residual that the closed-current projection cancels on $\mathrm{Crit}_n$ — the §7 carry-obstruction, whose order-4 instance — that $\Pi_\tau$ (projection onto the single-prime Cartan axes $A_2 \oplus A_3 \oplus A_5$) annihilates the mixed sector — is settled at this finite row: every cross-prime connected quartic vertex contains a mixed character (none of the 1078 is single-prime-only, by a mod-2 parity argument using $\dim A_2 = 1$), so all 914 three-prime vertices lie in $\ker \Pi_\tau$ while the single-axis $A_5$ sector survives carrying $\log 5$; no three-prime carry survives. As at cubic order this finite-row identity is the order-4 *instance* of the positivity program, not an unconditional proof of RH; the selection rule, the $\mathbb{Z}/30$ atlas, the single-axis von Mangoldt recovery, and the finite cross-prime cancellation are settled (certificates `empirical/verify_quartic_selection_z30.py`, `empirical/verify_recovery_current_z30.py`, and `empirical/verify_leakage_extinction_z30.py`; the first also resolves the finite-atlas computation named in Paper 21), while the exact mixed-residual cancellation on $\mathrm{Crit}_n$, at all orders, is the hard core (Lemma 8.1, Conjecture 5.2).

### 5.4 Reduction of the Full Limit

After Definition 5.1 and Theorems 5.1.2–5.2.1, the remaining gap splits into three concrete inputs.

1. **Euler-product/von Mangoldt recovery.**
   CRT factorization and the all-order support theorem reduce the prime side to showing that the primorial-smooth truncations converge, in the Weil test-function topology, to
   $$\sum_{p^k} \Lambda(p^k)\hat{f}(k\log p).$$
   This is a classical analytic-number-theory limit: a smooth-number cutoff, the prime number theorem in arithmetic progressions for the needed primorial residue classes, and a Tauberian passage to the completed explicit formula.

2. **Archimedean/spectral recovery.**
   The all-order Noether energy on the critical leaf is the proposed variational generator of the completed functional equation. Theorem 5.2.2 gives the archimedean gauge-lift criterion: under the Cartan floor, nonzero cubic $d$-symbol, closed-current compatibility, real-cone and Fisher-volume preservation, critical involution, and minimal positive-energy completion, the finite Cartan sector completes to $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$, and the Mellin character gives $\Gamma(s/2)\Gamma((s+1)/2)$. The remaining analytic work is to verify those lift axioms from the projective finite Noether tower rather than assume them.

3. **Projective closure / leakage extinction.**
   On the image of $\mathcal{R}_n$, Theorems 5.1.2 and 5.1.3 reduce leakage extinction to projective compatibility of the primorial restriction maps and the frame-price scaling condition
   $$\mu_n^\star(\mathcal{H}_n)\,\mathrm{polylog}(P_n) \to 0$$
   in the first-order-priced stratum, with exact vanishing in the hard packet-preserving stratum. Thus one must verify
   $$\Lambda_n(\mathcal{R}_n f) \to 0.$$
   A persistent off-critical component would define a nonzero limiting $\mathrm{Ext}_{CH}$ carry class in the projective primorial tower. Thus leakage extinction is equivalent to projective stabilization of the critical leaf, and the new local quantity that must be controlled along the squarefree/CRT primorial tower is $\mu_n^\star(\mathcal{H}_n)$ under the transported packet-escape normalization.

Together these three inputs are exactly Conjecture 5.2. No falsifying evidence appears in the finite rows currently inspected; the conjecture has been reduced to two classical finite-place analytic inputs, the Archimedean Noether Limit, and the squarefree/CRT verification of the projective-closure hypotheses.

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## 6. Conditional RH Theorem

**Theorem 6.1 (Conditional Primorial Noether proof of RH).**
Assume the Primorial Noether-Weil Identity. Then the Riemann Hypothesis is true.

*Proof.*
For $f \in \mathcal{S}_0$, positivity of the finite Noether form gives
$$\mathcal{Q}_n(\mathcal{R}_n f) \geq 0$$

for every $n$. This is a bounded-below statement for the finite information-geometric Noether/leakage form, not a physical energy condition. Taking the limit supplied by Conjecture 5.2 gives

$$\mathcal{W}_\zeta(f * f^*) \geq 0.$$

Since $\mathcal{S}_0$ is dense in the relevant Weil test-function space and the completed explicit-formula quadratic form is continuous under the chosen topology, positivity extends to the full Weil class. By Weil's positivity criterion, this is equivalent to all nontrivial zeros of the completed zeta function lying on $\mathrm{Re}(s) = 1/2$. Hence RH holds. □

This theorem is deliberately modest in one sense and maximal in another. It does not hide the hard analytic step. But it also says exactly what must be proved: identify the limiting all-order Noether energy on the primorial critical leaf with the Weil explicit-formula form.

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## 7. Why Quartic and Quintic Matter

The first finite tests are not arbitrary.

At quartic order the canonical row is

$$R_3 = \mathbb{Z}/30\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}.$$

This is the first row where a genuine three-prime conductor class can exist. It asks whether $2 \cdot 3 \cdot 5$ contains a carry obstruction that is not visible on the pairwise faces $6, 10, 15$. The order-4 atlas of §5.3 answers the existence half: 914 of the 1082 connected quartic vertices span all three primes and are reducible to no pairwise face — the non-pairwise content is present and counted. The first cumulant order to separate this content from the Gaussian (Wick-pairing) floor is precisely order 4, which is why this is the decisive row.

At quintic order the canonical row is

$$R_4 = \mathbb{Z}/210\mathbb{Z}.$$

Now the question is coherence among triple faces:

$$30, \quad 42, \quad 70, \quad 105.$$

Quartic order detects the first non-pairwise obstruction. Quintic order tests whether those non-pairwise obstructions compose coherently across the natural-number divisor lattice.

Both resolve through one observation, valid at every order and every primorial row. A single-prime (Cartan) character on axis $p$ is $\equiv 0 \pmod q$ for every other prime $q \mid P_n$; so were a connected vertex (sum $\equiv 0$, no proper subset $\equiv 0$) built from single-prime characters alone and spanning $\geq 2$ primes, the characters on any one axis would already form a proper subset summing to $0 \pmod{P_n}$ — the full sum is $0 \bmod p$, and they are 0 on every other axis — contradicting connectivity. Hence **no connected single-prime vertex spans $\geq 2$ primes**, and the closed-leaf projection $\Pi_\tau$ onto the single-prime Cartan axes annihilates the entire cross-prime sector — at orders $3, 4, 5, \ldots$ on $\mathbb{Z}/6, \mathbb{Z}/30, \mathbb{Z}/210, \ldots$ alike (certificate `empirical/verify_quintic_z210.py`, spot-checked through $\mathbb{Z}/210$). The non-pairwise obstructions thus compose *coherently by cancellation*: every cross-prime vertex carries $\Lambda = 0$ and vanishes on $\mathrm{Crit}_n$, with no surviving four-prime (or higher) carry; the new prime $p = 7$ enters only through the single-axis packet $A_7$ ($\Lambda(7) = \log 7$). The genuinely new *structural* feature at order 5 is its Wick subtraction — the 3+2 partition, a Gaussian pair times an

Amari–Chentsov cubic triple (the first order whose disconnected part carries the order-3 content). What this finite, all-rows argument does *not* settle is the uniform $n \to \infty$ limit (Conjecture 5.2): the cross-prime carry is removed at every finite row, leaving that limit as the sole obstruction.

Thus the primorial tower is not a decorative indexing device. It is the finite skeleton of the all-order prime interaction problem.

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## 8. What Remains

The conjecture is now cornered into four named obligations: the two finite-place analytic gaps, the squarefree/CRT verification of projective closure, and the Archimedean Noether Limit.

**Top task: Primorial-smooth von Mangoldt recovery.**
Prove the following analytic lemma with explicit cutoff choices.

**Lemma 8.1 (target).** For every $f \in \mathcal{S}_0$, there are cutoffs

$$p_n \to \infty, \qquad P_n = p_n\#, \qquad K_n \to \infty, \qquad M_n \to \infty$$

such that the finite Hayes-Mobius/CRT observable extracted from the recovery current in Definition 5.1 satisfies, locally uniformly for $f$ in bounded sets of the chosen Weil test class,

$$\mathcal{L}_n(f) = \sum_{\eta \in \widehat{R}_n} \hat{h}_n^f(\eta)\, \Phi_{\eta,n}(f), \qquad \lim_{n\to\infty} \mathcal{L}_n(f) = \sum_p \sum_{r\geq 1} (\log p)\, \hat{f}(r \log p).$$

For the completed Weil form, the same recovery must hold for the reflected test function, giving the symmetric finite-place distribution

$$\sum_p \sum_{r\geq 1} (\log p)(\hat{f}(r\log p) + \hat{f}(-r\log p)).$$

Here $\Phi_{\eta,n}$ is not an informal placeholder: it is the finite character observable induced by the order-truncated Hayes-Mobius coefficient vector after closed-sector projection and single-axis extraction, as made explicit below; along the diagonal, $\Phi_{\eta,n} = \Phi_{\eta,n}^{(K_n)}$.

This is the highest-priority task because it is the place where the primorial finite models must recover the actual natural-number Euler data. It should be attacked as a classical analytic-number-theory limit: smooth-number approximations, a Tauberian passage, prime number theorem input, and uniformity conditions chosen so that primorial residue classes stay inside known ranges.

The following reduction is the current rigorous proof plan. It does not close Lemma 8.1; it isolates exactly what must be proved and which parts are classical analytic number theory.

**Conventions for Section 8.**
Fix the completed Weil test-function topology once and for all. For $f \in \mathcal{S}_0$, write

$$g_f(t) = f(e^t), \qquad \hat{f}(u) = \int_{\mathbb{R}} g_f(t) e^{-iut}\, dt,$$

with the harmless $2\pi$-normalization in Definition 5.1 absorbed into the calibration between the finite Fourier datum and this log-Fourier transform. The dense working class $\mathcal{S}_0^{\exp}$ below is equipped with

seminorms controlling $e^{(1+\delta)|u|}\partial_u^j \widehat{f}(u)$ on bounded sets. This convention is needed so that the target von Mangoldt distribution is absolutely convergent before the final density passage.

For an occupation cutoff $K$, define the finite scalar observable

$$\mathcal{L}_{n,K}(f) := \mathrm{VM}_{n,K}\Big(\Pi^{\Lambda}_{\mathrm{Crit}_n}(\mathsf{C}_k^{(n)}(\hat{h}_n^{\,f}))_{2\le k\le K}\Big).$$

Here $\mathrm{VM}_{n,K}$ is the single-axis von-Mangoldt extractor: it applies the divisor-lattice primitive-support projector, keeps the closed-current single-prime-axis occupation components $re_p$, weights them by the Hayes-Mobius logarithmic coefficient, and discards mixed-axis packets. Equivalently, after expanding the polynomial current in one distinguished copy of $\hat{h}_n^{\,f}$,

$$\mathcal{L}_{n,K}(f) = \sum_{\eta\in\widehat{R}_n} \hat{h}_n^{\,f}(\eta)\,\Phi_{\eta,n}^{(K)}(f),$$

where $\Phi_{\eta,n}^{(K)}(f)$ contains the remaining convolution factors, the Hayes-Mobius coefficients, and the closed-sector projection. Lemma 8.1 is the assertion that $\mathcal{L}_{n,K_n}(f)$ has a diagonal limit as $n, K_n, M_n \to \infty$.

**Lemma 8.1a0 (CRT isolation of the finite observable).**
Let $R_n = \mathbb{Z}/P_n\mathbb{Z} \cong \prod_{p\le p_n} \mathbb{F}_p$. Additive character orthogonality, Ramanujan sums, and Mobius inversion on the divisor lattice identify the primitive support projector used by $\mathrm{VM}_{n,K}$ with the CRT conductor packet projector

$$H_{R_n} = \bigoplus_{d\mid P_n,\ d>1} H_n^{[d]}.$$

After this identification, the character observables $\Phi_{\eta,n}^{(K)}$ admit an expansion into Dirichlet characters and imprimitive character factors modulo divisors $q \mid P_n$, with local Euler factors at primes $p \mid q$ separated from the coprime residue classes. This is finite algebra, not an analytic prime-number input.

*Proof.* Index $\widehat{R}_n \cong R_n$ by $\eta = (\eta_p)_{p\le p_n}$ in CRT coordinates; the conductor is $d(\eta) = \prod_{p:\eta_p\neq 0} p$, and $H_n^{[d]} = \mathrm{span}\{e_\eta : d(\eta) = d\}$ gives the orthogonal grading $H_{R_n} = \bigoplus_{d\mid P_n} H_n^{[d]}$, with $\Pi_d$ the projector onto $H_n^{[d]}$. For $e \mid P_n$ let $E_e$ be the average over the cyclic subgroup $\langle e\rangle \le R_n$; its range is the annihilator $\langle e\rangle^{\perp} = \{\eta : (P_n/e) \mid \eta\} = \{\eta : d(\eta) \mid e\}$, so

$$E_e = \sum_{d\mid e} \Pi_d.$$

The primitive-support projector used by $\mathrm{VM}_{n,K}$ is the Mobius transform of these Ramanujan averages over the divisor lattice; by Mobius inversion,

$$P_d^{\mathrm{VM}} := \sum_{e\mid d} \mu(d/e)\,E_e = \sum_{d'\mid d}\Big(\sum_{d'\mid e\mid d}\mu(d/e)\Big)\Pi_{d'} = \Pi_d,$$

using $\sum_{d'\mid e\mid d}\mu(d/e) = [\,d' = d\,]$. This is the asserted identification of the $\mathrm{VM}_{n,K}$ projector with the CRT packet projector. The per-axis averages $A_p$ (average over $\langle P_n/p\rangle$; the projector onto $\{\eta : \eta_p = 0\}$) commute, and

$$\Pi_d = \prod_{p\mid d}(I - A_p)\prod_{p\nmid d} A_p$$

factors as a tensor product over the CRT axes; hence $\operatorname{rank}\Pi_d = \varphi(d) = \prod_{p|d}(p-1)$, and the observable $\Phi^{(K)}_{\eta,n}$ on the packet $d(\eta) = d$ separates into primitive local factors at $p \mid d$ and coprime-trivial factors at $p \nmid d$ — the finite Euler product. The packets are stable under $\eta \mapsto -\eta$, so the identity descends to the realified $H_n^{[d]}$. □

**Settlement (joint-on-disjoint, [P]).** The certificate `empirical/verify_lemma_8_1a0_crt_isolation.py` builds $P_d^{\mathrm{VM}}$ (cyclic-subgroup averages + Mobius) and $\Pi_d$ (per-prime Sylow averages + products) by two independent code paths in exact `Fraction` arithmetic, and confirms — with no floating point in any decision — that $P_d^{\mathrm{VM}} = \Pi_d$ for every $d \mid P_n$ at $P_n \in \{2, 6, 30\}$; idempotence, mutual orthogonality, completeness $\sum_d \Pi_d = I$, and $\operatorname{rank}\Pi_d = \varphi(d)$; and the squarefree Euler factorization with $\varphi(d) = \prod_{p|d}(p-1)$. The pre-registered joint-on-disjoint branch is realized: the projector identity holds even on the non-squarefree probe $\mathbb{Z}/12$, where the prime-power conductors $d \in \{4, 12\}$ appear as single-axis occupations — exactly the squarefree/occupation firewall to Lemma 8.1a, which the present lemma does *not* cross. Transcript: `empirical/reports/lemma_8_1a0_crt_isolation.md`.

**Lemma 8.1a (occupation lift from squarefree axes).**
Let

$$G_f(u) = \hat{f}(u).$$

For fixed $n$ and occupation cutoff $K$, let $\mathcal{L}_{n,K}(f)$ be the finite scalar observable defined above. The required structural statement is

$$\mathcal{L}_{n,K}(f) = \sum_{\substack{p \le p_n \\ 1 \le r \le K}} (\log p)\, G_f(r \log p) + \mathcal{M}_{n,K}(f) + \mathcal{R}^{\mathrm{leak}}_{n,K}(f),$$

where:

1. $\mathcal{M}_{n,K}(f)$ is the mixed-axis contribution indexed by occupations involving at least two distinct prime axes;
2. $\mathcal{M}_{n,K}(f) = 0$ on $\mathrm{Crit}_n$, or more generally $|\mathcal{M}_{n,K}(f)| \le C_f \Lambda_n(J_n(f))$;
3. $\mathcal{R}^{\mathrm{leak}}_{n,K}(f)$ is the finite leakage/projection error, hence vanishes in the hard packet-preserving stratum and is $O_f(\mu_n^\star(\mathcal{H}_n)\,\mathrm{polylog}(P_n))$ in the first-order-priced stratum.

**Writing this out: the squarefree/occupation split.** The decomposition above has two structurally different parts, and only one is finite squarefree algebra.

*(Skeleton — finite algebra, [P].)* The mixed/single separation is the radical content of the additive convolution and is settled by Lemma 8.1a0. Let $h$ be supported on the single-prime-axis (conductor-$p$) packets of $R_n$. Then every label of the $k$-fold additive convolution $h^{*k}$ has conductor equal to the radical of its lit axes; a single-axis convolution stays on conductor $p$ (or collapses to 1); and the full conductor $P_n$ first appears at order $k = n$. Hence the mixed-axis term $\mathcal{M}_{n,K}$ is supported exactly on the composite ($\ge$ 2-prime) conductors and is separated from the single-axis terms by the projectors $\Pi_d$ of Lemma 8.1a0. This is the conductor-convolution support theorem of Paper 10 read through the now-proved packet projectors.

*(Occupation lift — not squarefree bookkeeping; external, [A].)* The single-axis sum $\sum_r (\log p)\, G_f(r \log p)$ is **not** a function of the squarefree additive data on $R_n$. The additive dual $\widehat{R}_n \cong \prod_p \mathbb{F}_p$ records only the sum modulo $p$ on each axis; additive convolution on $\mathbb{F}_p$ wraps mod $p$, so occupation depth $r$ is invisible (depths $r$ and $r + p$ give the identical character), and the radical map $\mathbb{N}^{\{p \le p_n\}} \to \{\text{squarefree } d \mid P_n\}$, $k \mapsto \mathrm{rad}\,(\prod_p p^{k_p})$, forgets every depth — the prime-power tower $\{p^r : r \ge 1\}$ collapses to the single conductor $p$. So the von Mangoldt ladder (occupation $r$ at

position $r \log p$, weight $\Lambda(p^r) = \log p$) cannot be recovered from the squarefree skeleton: it lives on the occupation lattice $\mathbb{N}^{\{p \le p_n\}}$, which is infinite-to-one over the finite conductor lattice. This lift is exactly the "double cutoff" hypothesis of Theorem 5.2.1, not a consequence of squarefree bookkeeping. The phrase "remaining algebraic bookkeeping" therefore applies only to the skeleton; the occupation coefficient is an external (analytic, $K_n \to \infty$) obligation, the same gap named in the Claim-hygiene paragraph and carried with Lemma 8.1c.

**Settlement (joint-on-disjoint).** The certificate `empirical/verify_lemma_8_1a_occupation_lift.py` confirms, in exact arithmetic at $R \in \{6, 30\}$: the squarefree skeleton (conductors are radicals; full conductor first at order $n$; single-axis closure) — branch $\mathsf{K}$, [P]; and the occupation collapse (mod-$p$ wrap; each prime-power tower to one conductor; the occupation lattice infinite-to-one over the conductor lattice) — branch $\mathsf{L}$, localizing the occupation lift as external structure. Transcript: `empirical/reports/lemma_8_1a_occupation_lift.md`. The skeleton uses the conductor-convolution support theorem of Paper 10 and the Hayes-Mobius vectors of Paper 11; the occupation lift is the residual carried to Theorem 5.2.1 and Lemma 8.1c.

**Lemma 8.1b (double cutoff tail).**
Work first on a dense test class $\mathcal{S}_0^{\exp} \subset \mathcal{S}_0$ for which

$$\sum_p \sum_{r \ge 1} (\log p)\, |G_f(r \log p)| < \infty.$$

This is automatic if $G_f$ has compact support in the logarithmic variable; it also follows from a standard exponential log-side decay hypothesis, for example $e^{(1+\delta)|u|} G_f(u) \in L^1(du)$ with enough regularity for partial summation. Then for every cofinal diagonal

$$p_n \to \infty, \qquad K_n \to \infty, \qquad M_n \to \infty$$

such that

$$T_n(f) := \sum_{p > p_n \text{ or } r > K_n \text{ or } p^r > M_n} (\log p)\, |G_f(r \log p)| \longrightarrow 0,$$

one has

$$\sum_{\substack{p \le p_n \\ 1 \le r \le K_n \\ p^r \le M_n}} (\log p)\, G_f(r \log p) \longrightarrow \sum_p \sum_{r \ge 1} (\log p)\, G_f(r \log p).$$

Thus the axis cutoff, occupation cutoff, and Dirichlet cutoff commute on $\mathcal{S}_0^{\exp}$. This is an elementary dominated-convergence statement once the test space has enough log-side decay; without that decay, the admissible Weil test space must be narrowed or a separate Tauberian estimate must be supplied.

**Lemma 8.1c (primorial character uniformity).**
Let $q = P_n$. Let $\mathcal{X}_n$ be the finite set of Dirichlet characters and imprimitive character factors obtained by expanding the CRT character observables $\Phi_{\eta,n}^{(K)}$ into multiplicative characters modulo divisors of $q$, with the local Euler factors at primes $p \mid q$ separated from the coprime residue classes. A sufficient analytic input is the weighted character prime number theorem

$$\max_{\chi \in \mathcal{X}_n} \left| \sum_{m \le X} \Lambda(m) \chi(m) G_f(\log m) - \delta_{\chi = \chi_0} \int_1^X G_f(\log t)\, dt \right| \le X\, E_n(f),$$

uniformly on the $X$-ranges used by the smooth partition of the chosen Dirichlet cutoff, with $E_n(f) \to 0$ after the normalization used in the finite character observable. By partial summation

it is enough to prove the corresponding estimate for $\psi(X,\chi)$. Blocks below the uniform asymptotic range must be treated separately: either they are already exact finite occupation terms from Lemma 8.1a, or they must be negligible by the tail estimate in Lemma 8.1b.

There are two clean routes:

1. **Siegel-Walfisz-safe diagonal.** If $M_n$ is allowed to be very large, choose, for example,

$$M_n = \exp(P_n^\beta)$$

with fixed $\beta > 0$. Since $P_n \le (\log M_n)^A$ for every $A > 1/\beta$, the classical Siegel-Walfisz theorem gives individual uniformity on every dyadic block whose scale $X$ also satisfies $P_n \le (\log X)^A$, for all moduli dividing $P_n$, with ineffective constants but enough convergence for an existence theorem after the low-scale blocks have been handled as above.
2. **Natural-scale diagonal.** If one wants $M_n$ only polynomial or mildly exponential in $P_n$, textbook Siegel-Walfisz no longer applies. One must prove either a Bombieri-Vinogradov/large-sieve selector theorem for the primorial subsequence $q = P_n$, or a structural primorial-moduli argument excluding exceptional character contributions for these CRT observables.

The first route is classical but uses huge cutoffs. The second route is the sharp analytic gap if the recovery map is required to live at natural finite row scale.

The classical inputs allowed in Lemma 8.1c are therefore exactly: CRT and finite character orthogonality; Ramanujan sums and Mobius inversion on the conductor/divisor lattice; the prime number theorem and the prime number theorem in arithmetic progressions; Siegel-Walfisz only on ranges where $q \mid P_n$ satisfies $q \le (\log X)^A$; Bombieri-Vinogradov or Barban-Davenport-Halberstam only when the Hayes-Mobius observable needs averaged control over its character family; the large sieve for mean-square control of character-expanded errors; and smooth-weight partial summation, Mellin inversion, Perron/Tauberian tools. If the proof keeps the $P_n$-smooth integer truncation rather than passing directly to prime-power occupations, a de Bruijn/Hildebrand-Tenenbaum smooth-number estimate must also be included.

**Lemma 8.1d (density and topology).**
The preceding estimates must be locally uniform on bounded sets of $\mathcal{S}_0^{\exp}$, and $\mathcal{S}_0^{\exp}$ must be dense in the completed Weil test-function topology used for $\mathcal{W}_\zeta$. Then the identity extends from $\mathcal{S}_0^{\exp}$ to the final test class.

**Proposition 8.2 (conditional proof of Lemma 8.1 from the sublemmas).**
Assume Lemma 8.1a0 and Lemmas 8.1a–8.1d. Choose a diagonal $(n, K_n, M_n)$ satisfying the tail condition of Lemma 8.1b and one of the uniformity alternatives in Lemma 8.1c. If Theorem 5.1.3 gives leakage extinction, then the Dirichlet side of the recovery current satisfies

$$\lim_{n\to\infty} \mathcal{L}_{n,K_n}(f) = \lim_{n\to\infty} \sum_{\eta\in\widehat{R}_n} \hat{h}_n^f(\eta)\Phi_{\eta,n}^{(K_n)}(f) = \sum_p \sum_{r\ge1} (\log p)\hat{f}(r\log p) = \sum_{p^r} \Lambda(p^r)\hat{f}(\log p^r)$$

for $f \in \mathcal{S}_0^{\exp}$, and hence for all $f \in \mathcal{S}_0$ after Lemma 8.1d. With the standard completed-zeta normalization, the same argument must be run for the reflected test function as well, producing the symmetric finite-place distribution

$$\sum_p \sum_{r\ge1} (\log p)(\hat{f}(r\log p) + \hat{f}(-r\log p)).$$

*Proof.* Lemma 8.1a0 makes the finite observable precise and expands its CRT character content into the required primitive-support and residue-class pieces. Lemma 8.1a then converts the closed Hayes-Mobius recovery current into the single-axis occupation sum, up to mixed terms and leakage terms. The mixed terms vanish on the closed sector, or are bounded by $\Lambda_n(J_n(f))$, and Theorem 5.1.3 sends the leakage contribution to zero. Lemma 8.1c identifies the finite CRT character observable with the same occupation-weighted von Mangoldt coefficients, with error $E_n(f) \to 0$. Lemma 8.1b then removes the finite axis, occupation, and Dirichlet cutoffs. Finally Lemma 8.1d extends the identity to the completed Weil test space. □

**Claim hygiene.** At the present stage, Lemma 8.1a0 is proved and grounded (the projector identification and Euler separation; certificate `empirical/verify_lemma_8_1a0_crt_isolation.py`), making $\Phi_{\eta,n}^{(K)}$ explicit on each conductor packet; Lemma 8.1a is now written out as a squarefree/occupation split (certificate `empirical/verify_lemma_8_1a_occupation_lift.py`) — its squarefree skeleton is finite algebra [P], while the occupation lift to the prime-power lattice is external structure (the Theorem 5.2.1 double cutoff), not squarefree bookkeeping; Lemma 8.1b is elementary once the test space is fixed; Lemma 8.1c is classical via the Siegel-Walfisz-safe diagonal but remains a genuine analytic gap at natural primorial scale; and Lemma 8.1d is a functional-analytic density check. The squarefree datum alone does not recover prime powers; the $K_n \to \infty$ occupation lift is a separate obligation. Bombieri-Vinogradov gives averaged control and cannot be cited as pointwise control for the single full modulus $P_n$ unless the observable has first been reduced to an averaged character family. The cutoff schedule should be uniform on bounded subsets of the chosen test space, not chosen separately for each isolated $f$, unless the final theorem is explicitly weakened to pointwise recovery. These statements address only the finite-place von Mangoldt side; they do not prove the Archimedean Noether Limit, Weil positivity, or RH.

**Second task: Projective critical-leaf closure.**
Verify the hypotheses of Theorem 5.1.3 in the squarefree/CRT primorial tower: projective compatibility of the restriction maps, lower semicontinuity of the leakage forms, and the frame-price scaling condition

$$\mu_n^\star(\mathcal{H}_n)\,\mathrm{polylog}(P_n) \to 0.$$

Definition 5.1.3a, Criterion 5.1.3b, and Target Lemma 5.1.3c now spell this out as a finite geometric verification package. The remaining work is:

1. prove that the normalized CRT restrictions on $H_n$, packet projectors, and $\mathcal{J}_n/\mathrm{Flat}_n$ are functorial;
2. prove recovery compatibility

$$\pi_{m,n} J_m(f) = J_n(f) + \mathrm{flat} + o(1)$$

   together with cutoff compatibility for $M_n$ and the squarefree Weil coefficients;
3. prove the leakage restriction inequality, lower semicontinuity, and closed-sector stability after quotienting flats;
4. verify that leakage detects exactly the nonclosed cross-packet component on Hayes-Mobius currents and that

$$\Lambda_n(J) \le C_f\,\mathrm{polylog}(P_n)\varepsilon_n(J)$$

   for the transported current-level packet-escape seminorm;
5. transport Paper 25's first-order price $N(Q)$ to $\varepsilon_n(J)$ and keep the admissible stratum explicit: first-order price $\mu N$ with $\mu \ge \mu_n^\star$, or hard $O_{\mathrm{pkt}}$;

6. prove the squarefree CRT-face $\mu^\star$ stability theorem for proper faces and separately prove the full-conductor/top-face scaling law

$$\mu_n^\star(\mathcal{H}_n)\,\mathrm{polylog}(P_n) \to 0$$

or impose hard packet preservation.
7. test the Diophantine frame layer of Paper 25: on one-generator cyclic faces, prove or refute a three-gap carry grammar for sorted leakage/carry defects after split and flat terms are quotiented, and compare the resulting gap-price frontier with $\mu^\star$ and held-out frame transfer.

These hypotheses imply

$$\Lambda_n(\mathcal{R}_n f) \to 0$$

and that the projective limit of $\mathrm{Crit}_n$ is exactly the closed current space whose archimedean completion supplies the gamma factors. This is the non-classical finite-geometric input from the simplicial-cyclotomic program. It uses the Noether closure of Paper 17, the all-order support filtration of Paper 10, and the local $\mathrm{Ext}_{CH}$ obstruction bookkeeping of Paper 11. At the present stage this is a theorem schema plus finite diagnostic evidence, not yet a proved projective-closure theorem in the full squarefree/CRT tower. The three-gap/continued-fraction layer is a new candidate mechanism for the frame-price scaling condition: it suggests that priced carry on cyclic faces should have low Diophantine complexity, but this has not yet been proved for the primorial CRT tower.

**Third task: Archimedean Noether Limit.**
Verify the archimedean lift axioms of Theorem 5.2.2 from the projective finite Noether tower. Target Package 5.2.3 now spells this out as a Bures/projective checklist: prove the $R = 6$ Cartan floor, identify the nonzero cubic $d$-symbol internally in the Bures-unitary algebra, eliminate residual leakage upstairs, preserve the real cone and Fisher/Bures volume, realize the critical involution as an actual representation symmetry, prove minimal positive-energy completion, and upgrade the Mellin integral to a representation-theoretic character calculation. The theorem proves the forced completion and the character calculation once those axioms are in place: closed currents plus the nonzero cubic $d$-symbol force the minimal $\mathfrak{su}(2)\oplus\mathfrak{su}(3)$ completion, and after Mellin transform and summation over the lifted weight lattice the representation character reproduces $\Gamma(s/2)\Gamma((s+1)/2)$ with the functional-equation symmetry $s \leftrightarrow 1-s$. The rank bookkeeping is also explicit: $\mathfrak{u}(1)^2$ is a Cartan floor inside a rank-three completed algebra, so the extra Cartan direction must be central, leakage-flat, normalization-invisible, or included in the character.

Once these tasks are done, Theorem 5.2.1 upgrades Conjecture 5.2 to a theorem.

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## 9. Finite Consistency Evidence

The first finite rows support the conjecture.

1. **Quadratic row $R = 2$.**
The recovery map reduces to the identity on the single Fisher conductor packet. The finite Noether form is exactly the centered simplex Gram and matches the quadratic term of the completed Weil form.

2. **Cubic row $R = 6$.**
Exact symbolic character enumeration gives the conductor packets $\{2\}$, $\{3\}$, and $\{6\}$. The

closed-current projection leaves only the two Cartan $\mathfrak{u}(1)$ generators. The cubic Amari-Chentsov correction reproduces the $p = 2, 3$ prime contribution, and the mixed $2 \cdot 3$ term vanishes on the closed leaf as required by the functional-equation side.

3. **Quartic row $R = 30$.**
   The quartic cumulant tensor $T^{(4)}$ produces a nonzero triple-mixing term supported on the full conductor 30, not reducible to the pairwise faces $6, 10, 15$. This is the expected first appearance of the $2 \cdot 3 \cdot 5$ interaction.

4. **Quartic projection check.**
   After critical-leaf projection, the non-closed components are killed. The resulting four-linear form on short prime-word test functions agrees structurally with the truncated Weil kernel, with numerical agreement to the reported truncation precision. The normalized leakage on the recovered current is at machine precision in the finite check.

5. **No falsifier.**
   No new qualitative obstruction appears at quartic order beyond the representation-theoretic Cartan truncation already isolated in Paper 17. The $R = 30$ row behaves as predicted by the cumulant-order/primorial-row correspondence.

These checks are not a proof of RH. They show that the explicit recovery maps survive the first nontrivial rows and that the remaining obstruction is not a finite low-order contradiction. The next deliverable is an archived, reproducible certificate for the $R = 30$ comparison, followed by Lemma 8.1.

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## 10. Relation to Earlier Papers

Paper 5 supplied the cautious RH interfaces and warned that an operator, determinant, trace formula, or limiting theorem is required.

Paper 10 supplied the all-order conductor-convolution support theorem.

Paper 11 supplied the quotient-character adjacency algebra and the Hayes-Mobius coefficient vectors.

Paper 17 supplied the action-energy distinction, the Noether source of the cubic conserved quantities, and the Constraint 3 structural closure lemma for squarefree two-prime cubic support under packet-local/packet-respect hypotheses.

Paper 20 identified the jerk as a projected energy defect, suggesting that higher cumulants are not optional correction terms but leakage/mass data.

Paper 21 identified the quartic row 30 as the next natural obstruction.

This paper adds the missing global indexing principle:

$$\text{all-order cumulant Noether theory} \quad \text{runs over} \quad P_n = p_n\#.$$

That is the bridge from primes as local axes to natural numbers as the full occupation lattice.

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## 11. Safe and Unsafe Statements

**Safe statement.**
The primorial tower gives a canonical finite-stage model for the interaction of the first $n$ prime axes. The all-order cumulant Noether energy on its critical leaf is a plausible finite source for the positivity in Weil's explicit formula. Its positivity should be read as the same boundedness-below requirement that appeared at order 4 as the simplex mass/projection-closure residual, now imposed at the limit of the cumulant tower. The recovery maps are now explicit, the identity is exact at quadratic and cubic truncation, and the first quartic primorial row supplies no falsifying evidence. If the Primorial Noether-Weil Identity is proved, RH follows.

**Unsafe statement.**
The current finite computations already prove RH, or the limiting Noether form is a physical gravitational energy functional.

They do not. The missing theorem is not cosmetic. It is precisely the analytic recovery of the completed zeta explicit formula from the limiting primorial Noether form, with the primorial-smooth von Mangoldt recovery lemma as the next top task.

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## Conclusion

The cubic Lagrangian density was not the end of the story. It was the first non-Gaussian signal in an all-order cumulant action. Once the tower is read through CRT conductors, Lagrangian order points naturally to the primorial row: quartic to 30, quintic to 210, and the all-order limit to the prime-exponent lattice of the natural numbers.

The best current route to RH is therefore a Noether positivity route. The recovery maps are explicit, the low-order identities match, and the quartic row shows positive structural agreement. What remains is no longer a vague bridge: prove the primorial-smooth von Mangoldt recovery lemma, verify the projective closure hypotheses of Theorem 5.1.3, and prove the Archimedean Noether Limit. Then Theorem 5.2.1 identifies the limiting Noether energy with the Weil explicit-formula quadratic form, positivity is inherited as the boundedness-below of the finite information-geometric action, and RH follows by Weil's criterion.

This paper does not complete that final analytic identification. It names it: the Primorial Noether-Weil Identity. Its next technical component is Lemma 8.1. Until that identity and the Archimedean limit are proved, the bridge remains conditional and internal to the simplicial-cyclotomic information geometry.

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**Data availability:** This paper is theoretical. The relevant finite ingredients are the all-order conductor-convolution expansions of Paper 10, the quotient-character adjacency algebra of Paper 11, the cubic action-energy theorem of Paper 17, and the critical-leaf finite audits recorded in the knowledge base. The $R = 6$ and $R = 30$ consistency checks described here should be archived as reproducible symbolic certificates in the empirical directory. The first such artifact is `empirical/reports/z30_primorial_symbolic_certificate.md`, generated by `empirical/z30_primorial_symbolic_certificate.py`. The CRT isolation of Lemma 8.1a0 is settled by `empirical/verify_lemma_8_1a0_crt_isolation.py` (transcript `empirical/reports/lemma_8_1a0_crt_isolation.md`), an exact-arithmetic certificate at $P_n \in \{2, 6, 30\}$

with a $\mathbb{Z}/12$ firewall probe.

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*Ambitious draft. The Riemann Hypothesis is not claimed as proved here; the paper isolates a conditional proof route whose next task is explicit, technical, and falsifiable.*